

ON COMPACTIFICATIONS AND STONE-ČECH THEORY

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ABSTRACT. This paper explores the Stone-Čech compactification as a versatile tool in topology, known for its ability to extend locally compact Hausdorff spaces to a maximal compact framework. Through a detailed analysis of its universal properties, this compactification is presented as essential for understanding the boundary behavior of continuous functions and the broader structure of topological spaces. Key topics include the compactification's role in mapping spaces through ultrafilters, its applications in functional analysis, algebra, and topological dynamics, and its connections to set theory and combinatorics. By examining the implications of the Stone-Čech boundary and its interaction with logical and algebraic structures, this work underscores the compactification's profound influence in both theoretical and applied mathematical contexts, offering a foundation for future explorations in topology and related fields.

Mathematics is the science of the infinite, its goal the symbolic comprehension of the infinite with human, that is finite, means.

Hermann Weyl

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2020 Mathematics Subject Classification. Primary 54D35 Extensions of spaces (compactifications, supercompactifications, completions, etc.); Secondary 54D80 Special constructions of topological spaces (spaces of ultrafilters, etc.); 54D40 Remainders in general topology.

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1 INTRODUCTION

The mathematical process known as compactification was first discovered nearly a century ago. This mathematical gem lies at the crossroads of topology, algebra, analysis, category theory, dynamical systems, mathematical physics, and continues to yield interesting open problems. Compactification and its subset Stone-Čech Compactification show up in various areas of mathematics and physics, thus can be viewed from different lenses. We will begin with an introduction on the origin of compactification.

The first appearance of compactification appeared in an article titled “Ueber die topologische Erweiterung von Räumen” or “On the Topological Extension of Spaces” by Andrey Tikhonov published in the 102nd volume of *Mathematical Annals* in 1930. [1] In his work, Tikhonov built on Alexandrov’s foundational research on compact and locally compact spaces to further develop metrization criteria. Alexandrov’s paper, titled “Über die Metrisation der im Kleinen kompakten topologischen Räume” (On the Metrization of Small-scale Compact Topological Spaces), dealt with the conditions under which a locally compact space could be metrizable. Tikhonov’s extension of this work helped formalize the broader criteria required for general spaces, leading to a complete metrization theorem. In this paper he also introduces completely regular spaces. He proved that every Hausdorff completely regular space has a Hausdorff compactification. In 1937, Eduard Čech introduced the notation βX for this compactification using Tikhonov’s technique in a paper published in the *Annals of Mathematics* called “On Bicomact Spaces.” [2] In the same year, Marshall Stone also introduced a different construction of the compactification in his paper “Applications of the theory of Boolean rings to general topology” published in the *Transactions of the American Mathematical Society*. [3]

Čech’s work, alongside Stone’s earlier research, created a robust dual perspective — an algebraic view from Stone and a topological view from Čech — that would ultimately culminate in what is now known as the Stone-Čech compactification.

Further development in the mid-20th century, particularly through the contributions of mathematicians like Leonard Gillman and Meyer Jerison, helped solidify the understanding and application of βX . In their seminal 1960 text *Rings of Continuous Functions*, Gillman and Jerison offered an in-depth exploration of the link between compactifications and ring theory, examining how $C(X)$, the ring of continuous functions on a space X , interacts with its compactifications. [4] Their work not only provided clarity on the algebraic properties associated with βX but also introduced techniques that remain fundamental in modern topology.

The Stone-Čech compactification has since evolved into a crucial tool in fields such as functional analysis, algebraic topology, and set theory. Applications range from the study of ultrafilters and nonstandard analysis to the understanding of maximal compactifications. Notably, Walter Rudin's work on extremally disconnected spaces in the late 1950s and early 1960s provided additional insights into βX by examining conditions under which certain topological properties can be preserved under compactification.¹

In this paper, we embark on a comprehensive exploration of the Stone-Čech compactification, a central construct in topology with far-reaching applications across mathematical disciplines. The Stone-Čech compactification, often denoted βX , offers a method for extending a locally compact Hausdorff space X into a compact space that uniquely maximizes certain properties while preserving continuity. This construct allows mathematicians to study spaces "at infinity" and provides a powerful framework for understanding the asymptotic behavior of continuous functions.

The paper begins with a set-theoretic foundation, essential for formalizing the compactification process. Here, we introduce core concepts, including Zermelo-Fraenkel set theory, the Axiom of Choice, and key set-theoretic principles such as cardinality and the distinction between countable and uncountable sets. These preliminaries provide the logical and structural tools necessary for subsequent discussions.

Following this, we present essential topological constructs such as bases, subbases, product spaces, and quotient spaces, each critical for grasping the Stone-Čech compactification's role in extending spaces. Within this context, we define and explore compact, Hausdorff, and regular spaces, highlighting how these properties interact with the compactification.

The next sections are dedicated to examining various types of compactifications, from one-point to N -point compactifications, building up to the Stone-Čech compactification. Here, we examine different constructions of βX , including both the unit interval and ultrafilter methods. Each construction approach illuminates unique aspects of βX , from its universal mapping properties to its role in linking points in X to points "at infinity."

In the applications section, we illustrate the utility of the Stone-Čech compactification across several mathematical areas. For instance, in functional analysis, we discuss how the compactification extends function spaces, offering insights into bounded functions and C^* -algebras. Algebraic structures, particularly in semi-groups, benefit from the compactification through the study of ideal elements and maximal extensions. We also examine applications to set theory and topological dynamics, where the compactification aids in analyzing ultrafilters and the behavior of sequences.

The paper concludes with open questions and potential research directions, highlighting unresolved aspects of Stone-Čech theory and compactifications. We outline challenges in combinatorial applications and offer perspectives on the compactification's potential to address problems in logic and mathematical physics. Through

¹In mathematical literature, the names of Russian mathematicians Andrey Nikolayevich Tikhonov and Pavel Sergeyevich Alexandrov are often rendered as "Tychonoff" and "Alexandroff," respectively. This romanization follows historical conventions in mathematical texts, particularly in Western publications. The "-off" spelling reflects older transliteration practices from Russian to German or English, preserving phonetic similarities, which continue to be recognized in academic references. The romanized names will be used going forward for consistency.

this study, we aim to deepen the reader's understanding of compactification processes and underscore the enduring relevance of the Stone-Čech compactification in modern mathematical research.

2 SET-THEORETIC PRELIMINARIES

2.1 Zermelo–Fraenkel Set Theory with the Axiom of Choice. Zermelo–Fraenkel set theory with the Axiom of Choice (ZFC) provides the foundation for much of modern mathematics, offering a structured framework that supports rigorous proofs, construction of mathematical objects, and analysis of mathematical structures. In ZFC, all mathematical concepts are reducible to set-theoretic terms, and all proofs are conducted within this structure, ensuring consistency, clarity, and logical coherence in mathematical work. Additionally, the Axiom of Choice (AC) and its equivalents, such as Zorn's Lemma and the Well-Ordering Theorem, are essential for certain powerful results in analysis, topology, algebra, and beyond. Here's a formal presentation of the ZFC axioms, Zorn's Lemma, the Well-Ordering Theorem, and the maximal principle in set-theoretic notation.

Definition 2.1.1 *Zermelo–Fraenkel Axioms with the Axiom of Choice (ZFC)*

Each axiom contains both the logical notation and common language description.

- (1) *Axiom of Extensionality*

$$\forall A \forall B (\forall x (x \in A \Leftrightarrow x \in B) \Rightarrow A = B)$$

Any two sets A, B are equal if they have the same elements.

- (2) *Axiom of Empty Set*

$$\exists A \forall x (x \notin A)$$

There exists a set A with no elements, denoted by $\emptyset$.

- (3) *Axiom of Pairing*

$$\forall A \forall B \exists C \forall x (x \in C \Leftrightarrow (x = A \vee x = B))$$

For any sets A and B , there exists a set C containing exactly A and B .

- (4) *Axiom of Union*

$$\forall A \exists B \forall x (x \in B \Leftrightarrow \exists C (x \in C \wedge C \in A))$$

For any set A , there exists a set B that is the union of all elements of A .

- (5) *Axiom of Power Set*

$$\forall A \exists P \forall B (B \in P \Leftrightarrow B \subseteq A)$$

For any set A , there exists a set $\mathcal{P}$ (the power set of A) containing all subsets of A .

- (6) *Axiom of Infinity*

$$\exists A (\emptyset \in A \wedge \forall x (x \in A \Rightarrow x \cup \{x\} \in A))$$

There exists an infinite set A that contains the empty set $\emptyset$ and is closed under the operation of taking a set x and forming $x \cup \{x\}$ (i.e., adding x to itself as an element).

- (7) *Axiom of Replacement*

$$\forall A \forall F (\exists B \forall y (y \in B \Leftrightarrow \exists x (x \in A \wedge y = F(x))))$$

Applying a class function F to every element of a set A produces a new set B whose elements are precisely the results of applying F to elements of A .

(8) *Axiom of Regularity (Foundation)*

$$\forall A (A \neq \emptyset \Rightarrow \exists x (x \in A \wedge A \cap x = \emptyset))$$

Every non-empty set A contains an element x that is disjoint from A .

(9) *Axiom of Choice (AC)*

$$\forall A (\emptyset \notin A \Rightarrow \exists f : A \Rightarrow \bigcup A \text{ such that } \forall B (B \in A \Rightarrow f(B) \in B))$$

Given any set A of non-empty sets, there exists a choice function f that selects one element from each set in A .

Definition 2.1.2 The *Cartesian product* $X \times Y$ between two sets X and Y is the set of all possible ordered pairs with the first element from X and the second element from Y : $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$.

Definition 2.1.3 Given any non-empty set, S , a binary relation R is simply a subset of the Cartesian product $S \times S$. We tend to write " sRt " instead of " $(s, t) \in R$ ".

Definition 2.1.4 Let S be a non-empty set. A binary relation $R = "\leq"$ on S is called a *partial ordering* if it satisfies, for $s, t, u \in S$

- (i) $s \leq s$ (reflexivity)
- (ii) $s \leq t, t \leq u \Rightarrow s \leq u$ (transitivity)
- (iii) $s \leq t, t \leq s \Rightarrow s = t$ (anti-symmetry)

We call the pair $(S, \leq)$ a *partially ordered set*, also known as a *poset*.

Definition 2.1.5 In $(S, \leq)$, a *chain* is any subset C for which any two elements are comparable, i.e. for any $s, t \in C$, either $s \leq t$ or $t \leq s$. If S itself is a chain in $(S, \leq)$, we say that $\leq$ is a *total ordering* on S . If A is any subset of S , an upper bound for A is any u in S for which $s \leq u$ for s in A .

Definition 2.1.6 A *well-ordering* is any ordering $\leq$ on S such that in any non-empty subset A there is a minimal element, i.e. a in A such that $a \leq s$ for s in A . A well-ordered set is totally ordered.

Equivalent Statements to the Axiom of Choice: The Axiom of Choice is equivalent to several key propositions that are widely used in various areas of mathematics.

Theorem 2.1.7 *The following statements are equivalent:*

- (i) *Axiom of Choice (AC): for every non-empty X , there is a choice function, i.e. $f : P(X) \setminus \{\emptyset\} \rightarrow X$ such that $f(A) \in A$ for each A .*
- (ii) *Hausdorff's Maximality Principle: in any partially ordered set $(S, \leq)$ there is a maximal chain, i.e. a chain M for which no $M \cup \{s\}$ is a chain for any s in $S \setminus M$.*
- (iii) *Zorn's Lemma: if in a partially ordered set $(S, \leq)$, each chain has an upper bound, then there is a maximal element m for S , i.e. $m \leq s$ implies $m = s$.*
- (iv) *Well-ordering Principle: any non-empty set S admits a well-ordering.*

The proof of this theorem is beyond the scope of this paper, but the details of the proof can be found in [5].

The equivalence between the Axiom of Choice and Zorn's Lemma provides essential flexibility in establishing maximal and complete structures in topology. This equivalence allows us to prove the existence of maximal filters, ultrafilters, and bases, which are crucial for constructing compactifications, like the Stone-Čech compactification which we will encounter later on.

Tychonoff's Theorem, which states that any product of compact spaces is compact, also depends on the Axiom of Choice, especially for infinite collections. This theorem ensures that compactness extends to product spaces, an important property when working with compactified spaces. Together, these principles support the construction of compact, complete, and extended topological spaces by ensuring maximality and continuity across various proofs.

2.2 Finite, Countable, and Uncountable Sets.

Definition 2.2.1 A set S is *finite* if it has a limited number of elements. Formally, S is finite if there exists a natural number $n \in \mathbb{N}$ and a bijective function $f : S \rightarrow \{1, 2, \dots, n\}$. The cardinality (or size) of a finite set S is n .

Definition 2.2.2 A set S is *countable* if its elements can be put into a one-to-one correspondence with the natural numbers $\mathbb{N}$, meaning there exists a bijective function $f : S \rightarrow \mathbb{N}$ or a bijective function with a finite subset of $\mathbb{N}$. Equivalently, S is countable if it is either finite or *countably infinite*, where a countably infinite set has the same cardinality as $\mathbb{N}$.

Example 2.2.3 Examples include $\mathbb{N}$ itself, $\mathbb{Z}$ (the set of all integers), and $\mathbb{Q}$ (the set of all rational numbers).

Definition 2.2.4 A set S is *uncountable* if it is not countable, meaning there is no bijection between S and $\mathbb{N}$. In other words, the elements of S cannot be listed in a sequence. The cardinality of an uncountable set is strictly greater than that of $\mathbb{N}$.

Example 2.2.5 An example of an uncountable set is $\mathbb{R}$, the set of all real numbers, as proven by Cantor's diagonal argument. Another example is any interval in $\mathbb{R}$, such as $[0, 1]$.

3 POINT-SET TOPOLOGY PRELIMINARIES

We will start with a section on point-set topology to get acquainted with the tools needed.

Definition 3.0.1 [6] A *topology* on a set X is a collection τ of subsets of X having the following properties:

- (1) X and $\emptyset$ are in τ
- (2) The union of any subcollection of τ is in τ .
- (3) The intersection of any finite subcollection of τ is in τ .

A set X for which a topology τ has been specified is called a *topological space*. We will denote X and Y as topological space as we progress.

Definition 3.0.2 Let (X, τ) be a topological space. The elements of τ are called the *open sets* of X .

Example 3.0.3 We can define a topology τ on a set X as the collection of all subsets of X . This topology is called the *discrete topology* on X .

Example 3.0.4 We can define a topology τ on a set X as the collection consisting of X and $\emptyset$. This topology is called the *indiscrete topology* on X .

Example 3.0.5 Define τ to be the collection of all subsets U of X such that $X \setminus U$ is finite or is all of X . Then τ defines a topology on X , called the *cofinite topology* on X .

Example 3.0.6 The singleton space. A topology on a singleton set (one-point set) is a collection of subsets of the singleton set which contains the empty set and the entire singleton set and is closed under unions and finite intersections. Let $X = \{a\}$, then the only possible topology on X is $\tau = \{\emptyset, \{a\}\}$.

Example 3.0.7 A finite set is a topological space with only a finite number of elements. For example, a set with only two elements admits four distinct topologies. Let $X = \{a, b\}$ be a set with 2 elements. Then,

- (1) $\tau_1 = \{\emptyset, \{a, b\}\}$ (the indiscrete topology)
- (2) $\tau_2 = \{\emptyset, \{a\}, \{a, b\}\}$
- (3) $\tau_3 = \{\emptyset, \{b\}, \{a, b\}\}$
- (4) $\tau_4 = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$ (the discrete topology which is also a cofinite topology since X is finite)

Definition 3.0.8 A subset S of a topological space X is said to be *closed* if the set $X \setminus S$ is open.

Definition 3.0.9 Let (X, τ) be a topological space. Let $A \subseteq X$. A *neighborhood* of A is any subset of X containing an open set which itself contains A .

Definition 3.0.10 Given a subset $A \subseteq X$, the *interior* of A is defined as the union of all open sets contained in A , denoted by $\text{int } A$. The *closure* of A is defined as the intersection of all closed sets containing A , denoted by $\bar{A}$.

Definition 3.0.11 If $A \subseteq X$ and $x \in X$ then x is a *limit point* of A if it belongs to the closure of $A \setminus \{x\}$.

Theorem 3.0.12 Let A be a subset of the topological space X . Then $x \in \bar{A}$ if and only if every open set U containing x intersects A .

Proof: It is a statement of the form $P \Leftrightarrow Q$. Let us transform each implication to its contrapositive, thereby obtaining the logically equivalent statement $\neg P \Leftrightarrow \neg Q$. If x is not in $\bar{A}$, the set $U = X \setminus \bar{A}$ is an open set containing x that does not intersect A . Conversely, if there exists an open set U containing x which does not intersect A , then $X \setminus U$ is a closed set containing A . By definition of the closure $\bar{A}$, the set $X \setminus U$ must contain $\bar{A}$, therefore x cannot be in $\bar{A}$. $\square$

Theorem 3.0.13 Let $A \subseteq X$ and let A' be the set of all limit points of A . Then

$$\bar{A} = A \cup A'.$$

Proof: If $x \in A'$, every neighborhood of x intersects A (in a point different from x). Therefore, by Theorem 2.1, x belongs to $\bar{A}$. Hence, $A' \subseteq \bar{A}$. Since by definition $A \subseteq \bar{A}$, it follows that $A \cup A' \subseteq \bar{A}$. To show the reverse inclusion, we let x be a point of $\bar{A}$ and show that $x \in A \cup A'$. If x happens to lie in A , we know $x \in A \cup A'$. Suppose that x does not lie in A . Since $x \in \bar{A}$, we know that every neighborhood U of x intersects A . Because $x \notin A$, the set U must intersect A in a point different from x . Thus, $x \in A'$, so that $x \in A \cup A'$ as desired. $\square$

Theorem 3.0.14 *A subset of a topological space is closed if and only if it contains all its limit points.*

Proof: Assume A is closed. By definition, this means that $X \setminus A$ is open. Suppose, for the sake of contradiction, that there exists a limit point x of A such that $x \notin A$. Since $x \in X \setminus A$ and $X \setminus A$ is open, there exists an open neighborhood U of x with $U \subseteq X \setminus A$. However, being a limit point of A means that every open neighborhood of x must contain a point of A other than x itself. The fact that $U \subseteq X \setminus A$ implies U does not intersect A at all, which is a contradiction. Thus, no such x exists, and hence A contains all of its limit points.

Now assume A contains all of its limit points. We will show that A is closed. Let $x \in X \setminus A$. Since A contains all its limit points, x cannot be a limit point of A . Thus, there exists an open neighborhood U of x that does not intersect A at all. This means $U \subseteq X \setminus A$. Since this holds for any $x \in X \setminus A$, we conclude that $X \setminus A$ is open, which implies A is closed. $\square$

Definition 3.0.15 A subset A of a space X is said to be *dense* in X if $\bar{A} = X$.

Definition 3.0.16 If $f : A \rightarrow B$ and A_0 is subset of A , we define the *restriction* of f to A_0 to be the function mapping A_0 into B whose rule is $(a, f(a))$ such that $a \in A_0$ which is denoted by $f|_{A_0}$.

Definition 3.0.17 A function $f : A \rightarrow B$ is said to be *injective* (or *one-to-one*) if for each pair of distinct points of A , their images under f are distinct.

Definition 3.0.18 A function $f : A \rightarrow B$ is said to be *surjective* (or f is said to map A *onto* B) if every element of B is the image of some element of A under the function f .

Definition 3.0.19 If a function $f : A \rightarrow B$ is both injective and surjective, it is said to be *bijective* (or a *one-to-one correspondence*).

Definition 3.0.20 Let X and Y be topological spaces. A function $f : X \rightarrow Y$ is said to be *continuous* if for each open subset V of Y , the set $f^{-1}(V)$ is an open subset of X .

Definition 3.0.21 Let X and Y be two topological spaces with the mapping $f : X \rightarrow Y$ between them. If f is a continuous bijection and the function $f^{-1} : Y \rightarrow X$ is continuous as well, we call f a *homeomorphism*.

Definition 3.0.22 Let X and Y be topological spaces that admit a continuous one-to-one mapping $f : X \rightarrow Y$. Define $Z \subseteq Y$ as the image of X under f . Then we may use Z to define a new bijective mapping $g : X \rightarrow Z$. If g is a homeomorphism then f is an *imbedding* of X in Y .

Definition 3.0.23 Let X and Y be topological spaces. Let $f : X \rightarrow Y$ be a mapping. Then f is an *open imbedding* if and only if f is an imbedding and its image $f(X)$ is open in Y .

Lemma 3.0.24 *Let X and Y be topological spaces. let $f : X \rightarrow Y$. Then the following are equivalent:*

- (1) f is continuous.
- (2) For every subset A of X , one has $f(\bar{A}) \subset \overline{f(A)}$.

- (3) For every closed subset B of Y , the set $f^{-1}(B)$ is closed in X .
 (4) For each $x \in X$ and each neighborhood V of $f(x)$, there is a neighborhood U of x such that $f(U) \subset V$.

Proof: To begin, we assume that f is continuous and show that this implies statement (2). Let $A \subset X$ and let $\bar{A}$ denote the closure of A in X . By definition, $\bar{A}$ is the set of all points $x \in X$ such that every neighborhood of x intersects A non-trivially. For any point $y \in f(\bar{A})$, we have $y = f(x)$ for some $x \in \bar{A}$. Since f is continuous, for any open neighborhood $V \subset Y$ of y , the preimage $f^{-1}(V)$ is an open neighborhood of x in X . By the definition of closure, every neighborhood of x intersects A , so $f^{-1}(V) \cap A \neq \emptyset$. Thus, $V \cap f(A) \neq \emptyset$ for every neighborhood V of y , meaning $y \in \overline{f(A)}$. Hence, $f(\bar{A}) \subset \overline{f(A)}$, which establishes statement (2).

Next, assume statement (2) holds and show that it implies statement (3). Suppose $B \subset Y$ is closed. Then $Y \setminus B$ is open in Y , and we aim to show that $f^{-1}(B) \subset X$ is closed in X . Note that $f^{-1}(B) = \{x \in X : f(x) \in B\}$. Let $\overline{f^{-1}(B)}$ denote the closure of $f^{-1}(B)$ in X , which is the smallest closed subset of X containing $f^{-1}(B)$. By assumption, $f(\overline{f^{-1}(B)}) \subset \overline{f(f^{-1}(B))} \subset B$ since B is closed. Therefore, $\overline{f^{-1}(B)} \subset f^{-1}(B)$, so $f^{-1}(B)$ is closed in X , as required by statement (3).

Now, assume statement (3) and prove it implies statement (4). Let $x \in X$ and let $V \subset Y$ be a neighborhood of $f(x)$. We need to show there exists a neighborhood U of x in X such that $f(U) \subset V$. Since V is a neighborhood of $f(x)$, the complement $Y \setminus V$ is closed in Y . By assumption, the preimage $f^{-1}(Y \setminus V)$ is closed in X . Define $U = X \setminus f^{-1}(Y \setminus V)$, which is open in X because it is the complement of a closed set, and contains x since $f(x) \in V$. By construction, $f(U) \subset V$, satisfying statement (4).

Finally, assume statement (4) holds and show it implies continuity of f , thereby proving statement (1). Let $W \subset Y$ be open, and consider the preimage $f^{-1}(W) \subset X$. For each $x \in f^{-1}(W)$, we have $f(x) \in W$. By assumption, for each $x \in f^{-1}(W)$, there exists a neighborhood U_x of x in X such that $f(U_x) \subset W$. The collection of these neighborhoods $\{U_x : x \in f^{-1}(W)\}$ covers $f^{-1}(W)$, and each $U_x \subset f^{-1}(W)$ since $f(U_x) \subset W$. Therefore, $f^{-1}(W)$ is open in X as it is the union of open sets U_x , which shows that f is continuous. Thus, all four statements are equivalent. $\square$

Theorem 3.0.25 *Real polynomial functions are continuous on the interval $\mathbb{R} = (-\infty, \infty)$.*

Proof: Let f_n be an arbitrary real polynomial function of degree n . We proceed by induction. For all $n \in \mathbb{Z}^+$, let $P(n)$ be the proposition: f_n is continuous on $\mathbb{R}$. $P(0)$ is the case f_0 , where f_0 is of zero degree. Such a real polynomial function is a constant function. Since a constant real function is continuous, f_0 is continuous on $\mathbb{R}$. Thus $P(0)$ holds. Thus $P(1)$ holds. This is the basis for the induction. Next, it need to be shown that if $P(k)$ is true, where $k \geq 0$, then $P(k+1)$ is also true. Let the induction step be for all $x \in \mathbb{R}$ such that $f_{k+1}(x) = a_{k+1}x^{k+1} + f_k(x)$ for some real polynomial function of degree k , where $a_{k+1} \in \mathbb{R}$. From the induction hypothesis f_k is continuous on $\mathbb{R}$ and a real power function for a positive power is continuous thus f_{k+1} is continuous. So $P(k) \rightarrow P(k+1)$ by principle of mathematical induction. Therefore, for all $n \in \mathbb{Z}^+$, f_n is continuous on $\mathbb{R}$. $\square$

3.1 Basis, Subbasis and Subspaces.

Definition 3.1.1 Let X be a topological space. A *basis* for a topology on X is a collection $\mathcal{B}$ of subsets of X (called *basis elements*) such that

- (1) For each $x \in X$, there is at least one basis element B containing x .
- (2) If x belongs to the intersection of two basis elements B_1 and B_2 , then there is a basis element B_3 containing x such that $B_3 \subset B_1 \cap B_2$.

If $\mathcal{B}$ satisfies these two conditions, then we define the *topology τ generated by $\mathcal{B}$* as: A subset U of X is said to be open in X (that is, to be an element of τ) if for each $x \in U$, there is a basis element $B \in \mathcal{B}$ such that $x \in B$ and $B \subset U$.

Example 3.1.2 (*The standard topology of $\mathbb{R}$*) Let $\mathbb{R}$ be the set of real numbers. Let $\mathcal{B}$ be the collection of all open intervals: $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$. Then $\mathcal{B}$ is a basis of a topology and the topology generated by $\mathcal{B}$ is called the *standard topology* of $\mathbb{R}$.

Theorem 3.1.3 *Supposing the topology of X is given by a basis, then $x \in \bar{A}$ if and only if every basis element B containing x intersects A .*

If every open set containing x intersect A , so does every basis element B containing x , because B is an open set. Conversely, if every basis element containing x intersects A , so does every open set U containing x , because U contains a basis element that contains x and intersects A .

Definition 3.1.4 Let (X, τ) be a topological space. A collection $S \subseteq \tau$ is called a *subbasis* for τ if the following conditions hold:

- (1) The union of all sets in S is X .
- (2) The collection of all finite intersections of elements of S forms a basis for τ .

In other words, if you take every possible finite intersection of sets from S , these intersections produce a basis whose generated topology is exactly τ .

Definition 3.1.5 Let X be a topological space with topology τ . If Y is a subset of X , then the set $\tau_Y = \{Y \cap U \mid U \in \tau\}$ is a topology on Y called the *subspace topology*. With this topology, Y is called a *subspace* of X .

Example 3.1.6 The set $[0, 1]$ with the subspace topology of $\mathbb{R}$ is a topological space.

Example 3.1.7 The unit circle S^1 is defined by $S^1 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$. The circle S^1 is a topological space considered as a subspace of $\mathbb{R}^2$.

Example 3.1.8 In general, the n -dimensional sphere S^n is defined by $S^n = \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid x_1^2 + \dots + x_{n+1}^2 = 1\}$. It is a topological space considered as a subspace of $\mathbb{R}^{n+1}$.

3.2 Product, Box and Quotient Spaces.

Definition 3.2.1 A binary relation $\sim$ on a set X is said to be an *equivalence relation* if it is reflexive, symmetric and transitive; that is, for all $a, b, c \in X$,

- (1) $a \sim a$ (reflexive);
- (2) $a \sim b \Rightarrow b \sim a$ (symmetric);
- (3) $a \sim b$ and $b \sim c \Rightarrow a \sim c$ (transitive).

If $a, b \in X$, then a and b are said to be in the same *equivalence class* if $a \sim b$. In which the equivalence classes form a partition that is disjoint and whose union is the whole set.

Definition 3.2.2 Let $\pi_1 : X \times Y \rightarrow X$ be defined by the equation $\pi_1(x, y) = x$; let $\pi_2 : X \times Y \rightarrow Y$ be defined by the equation $\pi_2(x, y) = y$. The maps π_1 and π_2 are called the *projections* of $X \times Y$ onto its first and second factors, respectively.

Definition 3.2.3 Let X and Y be topological spaces. The *product topology* on $X \times Y$ is the topology that has as a basis all sets of the form $U \times V$, where U is an open subset of X and V is an open subset of Y . A Cartesian product equipped with a product topology is called a *product space*.

Definition 3.2.4 Suppose X is a topological space and $X_\alpha (\alpha \in A)$ is a family of topological spaces and that $f_\alpha : X \rightarrow X_\alpha$ for each α . The *evaluation map* defined by the family $\{f_\alpha : \alpha \in A\}$ is the function $E : X \rightarrow \prod_{\alpha \in A} X_\alpha$ such that $E(x)$ is the point in the product $\prod X_\alpha$ whose α^{th} coordinate is $f_\alpha(x)$. That is $E(x) = (f_\alpha(x))$.

Definition 3.2.5 Suppose X and $X_\alpha (\alpha \in A)$ are topological spaces and that $f_\alpha : X \rightarrow X_\alpha$. We say that the family $\{f_\alpha : \alpha \in A\}$ *separates points* if, for each pair of points $x \neq y \in X$, there exists an $\alpha \in A$ for which $f_\alpha(x) \neq f_\alpha(y)$.

Example 3.2.6 The *standard topology* of $\mathbb{R}^2$ is the product topology of $\mathbb{R} \times \mathbb{R}$ with the *standard topology*.

Definition 3.2.7 Let $\{X_\alpha\}_{\alpha \in A}$ be a family of topological spaces, where A is an index set. The *box topology* on the product space $X = \prod_{\alpha \in A} X_\alpha$ is the topology generated by the basis consisting of all sets of the form $\prod_{\alpha \in A} U_\alpha$, where $U_\alpha \subseteq X_\alpha$ is open in X_α for each $\alpha \in A$, and $U_\alpha = X_\alpha$ for all but finitely many α .

Definition 3.2.8 Let X and Y be topological spaces. Let $p : X \rightarrow Y$ be a surjective map. The map p is said to be a *quotient map* provided that a subset V of Y is open in Y if and only if $p^{-1}(V)$ is open in X .

Definition 3.2.9 Let X be a topological space, and let Y be a set. Suppose there exists a surjective map $q : X \rightarrow Y$. The *quotient topology* on Y is the topology where a subset $U \subseteq Y$ is declared open if and only if $q^{-1}(U)$ is open in X .

In other

Definition 3.2.10 Let X be a topological space and $\sim$ an equivalence relation on X . Let $X/\sim$ denote the set of equivalence classes in X , and let $\pi : X \rightarrow X/\sim$ be the natural projection sending each point to its equivalence class. Endowed with the quotient topology determined by π , the space $X/\sim$ is called the *quotient space* of X determined by the equivalence relation.

Example 3.2.11 Let X be the closed unit ball in $\mathbb{R}^2$, defined as

$$X = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}.$$

Consider the partition X^* of X , where each equivalence class is either a one-point set $\{(x, y)\}$ for points satisfying $x^2 + y^2 < 1$, or the boundary of the unit ball, denoted by

$$S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}.$$

This partition induces an equivalence relation $\sim$ on X , where $(x, y) \sim (x', y')$ if and only if either $(x, y) = (x', y')$ for points in the interior of the ball ($x^2 + y^2 < 1$), or both (x, y) and (x', y') lie on the boundary (S^1). Under this equivalence relation, all points on the boundary are treated as a single equivalence class, while points in the interior remain distinct.

3.3 Metric Spaces. We continue with the definition of a metric space which is a special type of topological space.

Definition 3.3.1 A metric d on a set X is a function $d : X \times X \rightarrow \mathbb{R}$ such that for all $x, y, z \in X$:

- (1) $d(x, y) \geq 0$ and $d(x, y) = 0$ if and only if $x = y$;
- (2) $d(x, y) = d(y, x)$;
- (3) $d(x, y) \leq d(x, z) + d(z, y)$.

A *metric space* (X, d) is a set X with a metric d defined on X .

Example 3.3.2 The set of *complex numbers* $\mathbb{C}$ with the metric $d(z, w) = |z - w|$.

Example 3.3.3 Any set X is metrizable by defining $d(x, y) = 1$ for any $x, y \in X$ with $x \neq y$, and $d(x, x) = 0$ for any $x \in X$.

Example 3.3.4 $\mathbb{R}^n$ is a metric space, with the standard Euclidean metric $d(x, y) = \|x - y\|$

Definition 3.3.5 Suppose (X, d) is a metric space and $a \in X$. For each $r > 0$, we define: the *open ball* in X centered at the point a and with a radius r to be the set $B_r(a) = \{x \in X \mid d(x, a) < r\}$.

As I will show, this definition aligns with the definition of an open set for topological spaces.

Definition 3.3.6 A subset U of a metric space (X, d) is called *open* if, for any point $x \in U$, there exists a real number $\epsilon > 0$ such that any point $y \in X$ satisfying $d(x, y) < \epsilon$ belongs to U . Equivalently, U is open if every point in U has a neighborhood contained in U .

Theorem 3.3.7 Every metric space is a topological space.

Proof: Let X be a set equipped with a metric d , and define τ as the collection of subsets $U \subseteq X$ such that for every point $x \in U$, there exists $\epsilon > 0$ with the open ball $B_d(x, \epsilon) = \{y \in X : d(x, y) < \epsilon\}$ contained in U . We claim that τ forms a topology on X .

First, the empty set $\emptyset$ and the entire set X are in τ because $\emptyset$ contains no points (satisfying the condition vacuously), and for any $x \in X$, $B_d(x, \epsilon) \subseteq X$ for any $\epsilon > 0$. Next, τ is closed under arbitrary unions. If $\{U_\alpha\}_{\alpha \in A}$ is a collection of sets in τ , then for any point $x \in \bigcup_{\alpha \in A} U_\alpha$, there exists some $\alpha_0 \in A$ such that $x \in U_{\alpha_0}$, and since $U_{\alpha_0} \in \tau$, there is an $\epsilon > 0$ with $B_d(x, \epsilon) \subseteq U_{\alpha_0} \subseteq \bigcup_{\alpha \in A} U_\alpha$. Therefore, $\bigcup_{\alpha \in A} U_\alpha \in \tau$. Finally, τ is closed under finite intersections. For sets $U_1, U_2, \dots, U_n \in \tau$, if $x \in \bigcap_{i=1}^n U_i$, then for each i , there exists $\epsilon_i > 0$ such that $B_d(x, \epsilon_i) \subseteq U_i$. Letting $\epsilon = \min\{\epsilon_1, \epsilon_2, \dots, \epsilon_n\}$, it follows that $B_d(x, \epsilon) \subseteq \bigcap_{i=1}^n U_i$. Thus, $\bigcap_{i=1}^n U_i \in \tau$.

Since τ satisfies the definition of a topology, it is called the topology induced by the metric d , and every metric space is therefore a topological space. $\square$

3.4 Fréchet and Hausdorff Spaces.

Definition 3.4.1 A topological space X is said to be a *Fréchet Space* if for any pair of distinct points of X , there exist two opens sets which each contain one point, but not the other, respectively. Formally, a topological space X is said to be a *Fréchet*

Space if for any $x, y \in X$, $x \neq y$ there exist open sets U and V such that $x \in U$, $y \notin U$ and $y \in V$, $x \notin V$.

Example 3.4.2 The real line $\mathbb{R}$ with the standard topology is a T_1 space. Since a T_1 space is a topological space in which every pair of distinct points can be separated by open sets, this property holds in $\mathbb{R}$ with the standard topology. Specifically, for any two distinct points $x, y \in \mathbb{R}$ where $x \neq y$, we can find open sets U and V in $\mathbb{R}$ such that, $x \in U$ and $y \notin U$, $y \in V$ and $x \notin V$. For instance, let $U = (x - \epsilon, x + \epsilon)$ and $V = (y - \delta, y + \delta)$ for some small $\epsilon > 0$ and $\delta > 0$ chosen such that the intervals do not overlap. Since $x \neq y$, such ϵ and δ can always be found, ensuring that U and V are disjoint open sets containing x and y , respectively.

Definition 3.4.3 A topological space X is called a *Hausdorff space* if for each pair x_1, x_2 of distinct points in X , there exist neighborhoods U_1 , and U_2 of x_1 and x_2 , respectively, that are disjoint.

Theorem 3.4.4 *Every finite subset in a Hausdorff space X is closed.*

Proof: It suffices to show that every one-point set $\{x_0\}$ is closed. If x is a point of X different from x_0 , then x and x_0 have disjoint neighborhoods U and V , respectively. Since U does not intersect $\{x_0\}$, the point x cannot belong to the closure of the set $\{x_0\}$. As a result, the closure of the set $\{x_0\}$ is $\{x_0\}$ itself, so that it is closed. $\square$

Definition 3.4.5 Let $(P, \leq)$ be a partially ordered set (poset), where P is a set and $\leq$ is a partial order on P . An *interval* in P is a subset $I \subseteq P$ with the property that for all $a, b \in I$ and any $c \in P$, if $a \leq c \leq b$, then $c \in I$.

There are several types of intervals defined based on the inclusion or exclusion of endpoints. A *closed interval* is given by $[a, b] = \{x \in P : a \leq x \leq b\}$, which is non-empty only if $a \leq b$. An *open interval* is defined as $(a, b) = \{x \in P : a < x < b\}$, where $a < x$ means $a \leq x$ and $a \neq x$, and $x < b$ means $x \leq b$ and $x \neq b$. *Half-open intervals* include $[a, b) = \{x \in P : a \leq x < b\}$ and $(a, b] = \{x \in P : a < x \leq b\}$. Additionally, *unbounded intervals* are defined as $(-\infty, b] = \{x \in P : x \leq b\}$, $[a, \infty) = \{x \in P : a \leq x\}$, $(-\infty, b) = \{x \in P : x < b\}$, and $(a, \infty) = \{x \in P : a < x\}$.

Definition 3.4.6 (*The order topology*) Let $(X, \leq)$ be a partially ordered set. Then the *order topology* on X is the topology generated by the basis $\mathcal{B}$ consisting of unions of sets of the form:

- (1) Open intervals of the form (a, b) with $a \leq b$ in X .
- (2) If X has a smallest element a_0 , then we also include half-open intervals $[a_0, b)$ with $a_0 \leq b$ in X .
- (3) If X has a largest element b_0 , then we also include half-open intervals of the form $(a, b_0]$ with $a \leq b_0$.

Theorem 3.4.7 *The product of two Hausdorff spaces is a Hausdorff space.*

Proof: Let X and Y be Hausdorff spaces, and let (x_1, y_1) and (x_2, y_2) be two distinct points in $X \times Y$. Since $(x_1, y_1) \neq (x_2, y_2)$, it follows that either $x_1 \neq x_2$ or $y_1 \neq y_2$.

Case 1: Suppose $x_1 \neq x_2$. Since X is Hausdorff, there exist disjoint open sets U_1 and U_2 in X such that $x_1 \in U_1$, $x_2 \in U_2$, and $U_1 \cap U_2 = \emptyset$. Then $U_1 \times Y$ and $U_2 \times Y$ are open in $X \times Y$. Moreover, $(x_1, y_1) \in U_1 \times Y$, $(x_2, y_2) \in U_2 \times Y$, and

$$(U_1 \times Y) \cap (U_2 \times Y) = (U_1 \cap U_2) \times Y = \emptyset \times Y = \emptyset.$$

Hence, (x_1, y_1) and (x_2, y_2) are separated.

Case 2: Suppose $y_1 \neq y_2$. By the Hausdorff property of Y , there exist disjoint open sets V_1 and V_2 in Y such that $y_1 \in V_1$, $y_2 \in V_2$, and $V_1 \cap V_2 = \emptyset$. Then $X \times V_1$ and $X \times V_2$ are open in $X \times Y$. Similarly, $(x_1, y_1) \in X \times V_1$, $(x_2, y_2) \in X \times V_2$, and

$$(X \times V_1) \cap (X \times V_2) = X \times (V_1 \cap V_2) = X \times \emptyset = \emptyset.$$

Thus, (x_1, y_1) and (x_2, y_2) are separated.

In both cases, $X \times Y$ satisfies the Hausdorff property. Therefore, $X \times Y$ is a Hausdorff space. $\square$

Theorem 3.4.8 *A subspace of a Hausdorff space is a Hausdorff space.*

Proof: Let A be a subspace of a Hausdorff space X and let a and b be two distinct in A . Since A is Hausdorff there exists two open sets U and V in X such that $a \in U$, $b \in V$ and $U \cap V = \emptyset$. Hence, $a \in A \cap U$, $b \in A \cap V$ and $(A \cap U) \cap (A \cap V) = (U \cap V) \cap A = \emptyset \cap A = \emptyset$. Since $A \cap U$ and $A \cap V$ are open in A , it follows that A is Hausdorff. $\square$

Lemma 3.4.9 *Let (X, d) be a metric space. Let $x, y \in X$ such that $x \neq y$. Then there exist disjoint open ϵ -balls $B_\epsilon(x)$ and $B_\epsilon(y)$ containing x and y respectively.*

Proof: Let $d(x, y) = 2\epsilon$. Let $B_\epsilon(x)$ and $B_\epsilon(y)$ denote the open ϵ -balls of x and y in M . From open balls whose distance between centers is twice radius are disjoint, $B_\epsilon(x)$ and $B_\epsilon(y)$ are disjoint open ϵ -balls. $\square$

Theorem 3.4.10 *Every finite point set in a Hausdorff space X is a closed.*

Proof: It suffices to show that every one-point set x_0 is closed. If x is a point of X different from x_0 , then x and x_0 have disjoint neighborhoods U and V , respectively. Since U does not intersect x_0 , the point x cannot belong to the closure of the set x_0 . As a result, the closure of the set x_0 is x_0 itself, so that it is closed. $\square$

3.5 Regular, Completely Regular and Normal Spaces.

Definition 3.5.1 Let X be a topological space. Suppose that one-point sets are closed in X . A topological space X is said to be *regular* if for each pair consisting of a point x and a closed set B disjoint from x , there exist disjoint open sets containing x and B , respectively.

Theorem 3.5.2 *Let X be a topological space. Suppose one-point sets in X are closed. X is regular if and only if given a point x of X and a neighborhood U of x , there is a neighborhood V of x such that $\bar{V} \subseteq U$.*

Proof: Suppose that X is regular, and suppose that the point x and the neighborhood U of x are given. Let $B = X \setminus U$ then B is a closed set. By hypothesis, there exist disjoint open sets V and W containing x and B respectively. The set $\bar{V}$ is disjoint from B since if $y \in B$, the set W is a neighborhood of y disjoint from V . Therefore, $\bar{V} \subseteq U$, as desired. Conversely, suppose the point x and the closed set B not containing x are given. Let $U = X \setminus B$. By hypothesis, there is a neighborhood V of x such that $\bar{V} \subseteq U$. The open sets V and $X \setminus \bar{V}$ are disjoint open sets containing x and B , respectively. Thus X is regular. $\square$

Theorem 3.5.3 *Metric spaces are regular spaces.*

Proof: Let (X, d) be a metric space. To show that metric spaces are regular, we start with the definitions. A topological space X is *regular* if, given any closed subset $C \subset X$ and any point $x \notin C$, there exist disjoint open sets U and V such that $x \in U$ and $C \subset V$. A space is said to satisfy the T_3 separation axiom if it is both T_1 (i.e., every single point set is closed) and regular. Since metric spaces are T_1 , it suffices to demonstrate that metric spaces are regular.

To prove regularity, let $C \subset X$ be a closed subset and $x \notin C$ a point in X but not in C . Since $x \notin C$ and C is closed, the distance $d(x, C) = \inf\{d(x, c) \mid c \in C\}$ is strictly positive. Define $\epsilon = \frac{d(x, C)}{2}$, so that for every $c \in C$, $d(x, c) > 2\epsilon$.

We construct the open sets $U = B(x, \epsilon) = \{y \in X \mid d(x, y) < \epsilon\}$, the open ball centered at x with radius ϵ , and $V = \bigcup_{c \in C} B(c, \epsilon) = \{y \in X \mid \exists c \in C \text{ such that } d(y, c) < \epsilon\}$. It remains to show that U and V are disjoint. For any $y \in U$, $d(x, y) < \epsilon$. For any $z \in V$, there exists a point $c \in C$ such that $d(z, c) < \epsilon$. Then, by the triangle inequality, $d(x, z) \geq d(x, c) - d(z, c) > 2\epsilon - \epsilon = \epsilon$. Therefore, $z \notin U$, and hence $U \cap V = \emptyset$.

Thus, we have constructed disjoint open sets U and V such that $x \in U$ and $C \subset V$, proving that X is regular. Therefore, metric spaces are regular spaces. $\square$

Definition 3.5.4 A topological space X is *Tychonoff* (completely regular) if singleton sets are closed in X and if for each point x_0 and each closed set A not containing x_0 , there is a continuous function $f : X \rightarrow [0, 1]$ such that $f(x_0) = 1$ and $f(A) = 0$.

Theorem 3.5.5 (Urysohn Imbedding Theorem) *Let X be a space in which singleton sets are closed. Suppose that $\{f_\alpha\}_{\alpha \in J}$ is an indexed family of continuous functions $f_\alpha : X \rightarrow \mathbb{R}$ satisfying the requirement that for each point x_0 of X and each neighborhood U of x_0 , there is an index $\alpha \in J$ such that f_α is positive at x_0 and vanishes outside U . Then the function $F : X \rightarrow \mathbb{R}^J$ defined by $F(x) = (f_\alpha(x))_{\alpha \in J}$ is an imbedding of X in $\mathbb{R}^J$. If f_α maps X into $[0, 1]$ for each α , then F embeds x in $[0, 1]^J$.*

Proof: To begin, we show that F is injective. Suppose $x, y \in X$ with $x \neq y$. Since X is a T_1 space, singleton sets are closed, and for any two distinct points x and y , there exists a neighborhood of x that does not contain y , and vice versa. By the assumption in the theorem, for each point $x_0 \in X$ and any neighborhood U of x_0 , there exists some index $\alpha \in J$ such that $f_\alpha(x_0) > 0$ and f_α vanishes outside U . Since $x \neq y$, we can choose a neighborhood U of x that does not contain y . Then there exists an index $\alpha \in J$ such that $f_\alpha(x) > 0$ and $f_\alpha(y) = 0$. Consequently, $F(x) \neq F(y)$, proving that F is injective.

Next, we verify that F is continuous. Since each f_α is continuous by assumption, the function F , which is a map into the product space $\mathbb{R}^J$ (or $[0, 1]^J$ if each f_α maps into $[0, 1]$), is also continuous in the product topology. Hence, F is continuous.

Finally, we show that F is a homeomorphism onto its image by proving that F is an open map onto its image. Let $V \subseteq X$ be open, and consider the image $F(V) \subseteq \mathbb{R}^J$. Since X is a T_1 space, for each point $x_0 \in V$, we can find an index $\alpha \in J$ such that $f_\alpha(x_0) > 0$ and f_α vanishes outside some neighborhood of x_0 . This condition ensures that small changes in x around x_0 lead to small changes in $F(x)$ in the product topology, so $F(V)$ is open in $F(X)$. Thus, F is an open map onto its image, establishing that F is a homeomorphism onto its image.

Since F is injective, continuous, and an open map onto its image, F is an embedding of X in $\mathbb{R}^J$ (or $[0, 1]^J$ if each f_α maps into $[0, 1]$). $\square$

Theorem 3.5.6 *Completely regular spaces are Hausdorff.*

Proof: Let X be a completely regular space. By definition, for any closed set $A \subset X$ and any point $p \in X \setminus A$, there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(p) = 0$ and $f(x) = 1$ for all $x \in A$.

To prove that X is Hausdorff, we need to show that any two distinct points in X can be separated by disjoint neighborhoods. In other words, for any two distinct points $p, q \in X$, there exist disjoint open sets U and V such that $p \in U$ and $q \in V$.

Suppose $p, q \in X$ and $p \neq q$. We aim to separate these points using disjoint open sets.

By the definition of complete regularity, since $p \neq q$, we can apply the property of completely regular spaces to the closed set $\{q\}$ and the point $p \notin \{q\}$. Therefore, there exists a continuous function $f : X \rightarrow [0, 1]$ such that:

$$f(p) = 0 \quad \text{and} \quad f(q) = 1.$$

Now, define the following open sets:

$$U = f^{-1}\left(\left[0, \frac{1}{2}\right)\right) \quad \text{and} \quad V = f^{-1}\left(\left(\frac{1}{2}, 1\right]\right).$$

These sets are open because f is continuous and the preimage of an open set under a continuous function is open.

Clearly, $p \in U$ (since $f(p) = 0$) and $q \in V$ (since $f(q) = 1$). Moreover, the sets U and V are disjoint, because the intervals $[0, \frac{1}{2})$ and $(\frac{1}{2}, 1]$ do not overlap.

Therefore, for any two distinct points $p, q \in X$, we have found disjoint open sets U and V such that $p \in U$ and $q \in V$. This proves that X is Hausdorff. $\square$

Theorem 3.5.7 *If X is a completely regular space and $Y \subset X$ is a subspace with the subspace topology, then Y is completely regular.*

Proof: Let $\{X_i\}_{i \in I}$ be Hausdorff spaces, and let $x = (x_i)$ and $y = (y_i)$ be two distinct points in $\prod_{i \in I} X_i$. Then there exists at least one index $j \in I$ such that $x_j \neq y_j$. Since X_j is Hausdorff, there exist disjoint open sets $U_j, V_j \subset X_j$ containing x_j and y_j , respectively. Define $U = \pi_j^{-1}(U_j)$ and $V = \pi_j^{-1}(V_j)$. Then U and V are open in the product topology, disjoint, and contain x and y , respectively. $\square$

Lemma 3.5.8 *The space $[0, 1]$ is completely regular.*

Proof: By example 2.30 we know $[0, 1]$ is a topological space. Next, note that any closed set $A \subseteq [0, 1]$ is compact because $[0, 1]$ is a compact space. This is a key property we will use to construct a separating function.

Let $x \in [0, 1] \setminus A$, i.e., x is a point outside the closed set A . Since A is closed and $[0, 1]$ is a Hausdorff space, there exist disjoint open sets separating x and A . Specifically, for each $y \in A$, there exist disjoint open sets U_y and V_y such that $x \in U_y$ and $y \in V_y$.

The collection of open sets $\{V_y : y \in A\}$ forms an open cover of A . Since A is compact, there is a finite subcover, say $V_{y_1}, V_{y_2}, \dots, V_{y_n}$, that covers A . Let the corresponding open sets $U_{y_1}, U_{y_2}, \dots, U_{y_n}$ be the ones containing x .

Using Urysohn's Lemma, since A is compact and separated from x , there exists a continuous function $f : [0, 1] \rightarrow [0, 1]$ such that:

- $f(x) = 0$
- $f(y) = 1$ for all $y \in A$

The function f separates the point x from the closed set A , and f is continuous. Therefore, the space $[0, 1]$ satisfies the condition of complete regularity: for every closed set $A \subseteq [0, 1]$ and point $x \notin A$, there exists a continuous function that separates them. Thus, $[0, 1]$ is a completely regular space. $\square$

Lemma 3.5.9 *Continuous functions separate points in a completely regular space.*

Proof: Let X be a completely regular space. By definition, for any closed set $A \subset X$ and any point $p \in X \setminus A$, there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(p) = 0$ and $f(x) = 1$ for all $x \in A$.

We need to show that for any two distinct points $p, q \in X$, there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(p) \neq f(q)$. This is equivalent to saying that continuous functions separate points.

Since $p \neq q$, we can consider the singleton sets $\{p\}$ and $\{q\}$, which are closed in a Hausdorff space (a completely regular space is Hausdorff).

By the definition of complete regularity, since $p \notin \{q\}$ (and similarly $q \notin \{p\}$), there exist continuous functions that separate these points from each other. Specifically, there exist continuous functions $f_1 : X \rightarrow [0, 1]$ such that:

$$f_1(p) = 0 \quad \text{and} \quad f_1(q) = 1,$$

and similarly, there exists a continuous function $f_2 : X \rightarrow [0, 1]$ such that:

$$f_2(p) = 1 \quad \text{and} \quad f_2(q) = 0.$$

Either one of these functions f_1 or f_2 separates p and q , as their values differ at the points p and q .

Therefore, we have found a continuous function $f : X \rightarrow [0, 1]$ such that $f(p) \neq f(q)$, which demonstrates that continuous functions separate points in a completely regular space. $\square$

Lemma 3.5.10 *The space $\mathbb{R}^J$ in the box topology is completely regular.*

Proof: To prove that $\mathbb{R}^J$ in the box topology is completely regular, we must show that for any closed subset $C \subseteq \mathbb{R}^J$ and any point $x \notin C$, there exists a continuous function $f : \mathbb{R}^J \rightarrow [0, 1]$ such that $f(x) = 0$ and $f(y) = 1$ for all $y \in C$.

In the box topology on $\mathbb{R}^J$, a basis consists of all products $\prod_{j \in J} U_j$, where each U_j is an open set in $\mathbb{R}$, and $U_j = \mathbb{R}$ for almost all j . A space X is completely regular if, for any closed subset $C \subseteq X$ and any point $x \notin C$, there exists a continuous function $f : X \rightarrow [0, 1]$ that separates x from C by satisfying $f(x) = 0$ and $f(y) = 1$ for all $y \in C$. Since $\mathbb{R}$ itself is completely regular, for any point $x_j \in \mathbb{R}$ and closed subset $C_j \subseteq \mathbb{R}$ such that $x_j \notin C_j$, we can find a continuous function $f_j : \mathbb{R} \rightarrow [0, 1]$ with $f_j(x_j) = 0$ and $f_j(y_j) = 1$ for all $y_j \in C_j$.

Now, consider a point $x = (x_j)_{j \in J} \in \mathbb{R}^J$ and a closed subset $C \subseteq \mathbb{R}^J$ such that $x \notin C$. For each $j \in J$, define $C_j = \{y_j \in \mathbb{R} : (y_j)_{j \in J} \in C\}$, noting that since C is closed in the box topology, each C_j is closed in $\mathbb{R}$, and $x_j \notin C_j$. For each $j \in J$,

let $f_j : \mathbb{R} \rightarrow [0, 1]$ be a continuous function that separates x_j from C_j by satisfying $f_j(x_j) = 0$ and $f_j(y_j) = 1$ for all $y_j \in C_j$. We then define $f : \mathbb{R}^J \rightarrow [0, 1]$ by

$$f((z_j)_{j \in J}) = \sup_{j \in J} f_j(z_j).$$

The function f is continuous in the box topology, as each f_j is continuous in the standard topology on $\mathbb{R}$, and the box topology on $\mathbb{R}^J$ ensures that taking the supremum over coordinates yields a continuous function. This function f separates x from C , since $f(x) = 0$ and $f(y) = 1$ for all $y \in C$. Hence, $\mathbb{R}^J$ in the box topology is completely regular. $\square$

Definition 3.5.11 A topological space X is called a *normal space* if it satisfies the following two conditions:

- (1) X is a Fréchet (T_1) space, meaning that for any two distinct points $x, y \in X$, there exist open sets U and V such that $x \in U$, $y \in V$, and neither contains the other point.
- (2) For any two *disjoint closed subsets* A and B of X , there exist *disjoint open sets* U and V such that $A \subseteq U$ and $B \subseteq V$.

3.6 Connected and Compact Spaces.

Definition 3.6.1 Let X be a topological space. A *separation* of X is a pair U, V of disjoint nonempty open subsets of X whose union is X .

Definition 3.6.2 A topological space X is *connected* if and only if the only subsets of X that are both open and closed in X are the empty set and X itself.

Definition 3.6.3 A topological space X is *totally disconnected* if its only connected subspaces are one-point (singleton) sets.

Definition 3.6.4 A collection $\mathcal{A}$ of subsets of a space X is said to *cover* X or to be a *covering* of X , if the union of elements of $\mathcal{A}$ is equal to X . $\mathcal{A}$ is called an *open covering* of X if its elements are open subsets of X .

Definition 3.6.5 A topological space X is said to be *compact* if every open cover of X has a finite subcover that also covers X .

We say X is *topologically locally compact* at a point x if there exists a compact subspace $C \subseteq X$ such that C contains a neighborhood of x .

If X is locally compact at each of its points, we say X is *locally compact*.

Definition 3.6.6 Let $\mathcal{F}$ be a collection of sets. Then $\mathcal{F}$ is said to have the *Finite Intersection Property* if for every finite subcollection $\{F_1, F_2, \dots, F_n\} \subseteq \mathcal{F}$, we have that $\bigcap_{i=1}^n F_i \neq \emptyset$.

Theorem 3.6.7 (Cantor Intersection Theorem) *Let S be a topological space. A decreasing nested sequence of non-empty compact, closed subsets of S has a non-empty intersection. In other words, if $(C_k)_{k \geq 0}$ is a sequence of non-empty compact, closed subsets of S such that*

$$C_0 \supset C_1 \supset C_2 \supset \dots \supset C_n \supset C_{n+1} \supset \dots,$$

then

$$\bigcap_{k=0}^{\infty} C_k \neq \emptyset.$$

The closedness condition may be omitted in situations where every compact subset of S is closed, for example, when S is Hausdorff.

Proof: Assume, by way of contradiction, that

$$\bigcap_{k=0}^{\infty} C_k = \emptyset.$$

For each k , define the set

$$U_k = C_0 \setminus C_k.$$

Since

$$\bigcup_{k=0}^{\infty} U_k = C_0 \setminus \left(\bigcap_{k=0}^{\infty} C_k \right) = C_0,$$

and by the assumption that $\bigcap_{k=0}^{\infty} C_k = \emptyset$, we have $\bigcup_{k=0}^{\infty} U_k = C_0$. Each C_k is closed relative to S , and thus closed relative to C_0 . Therefore, each U_k , as the complement of C_k in C_0 , is open relative to C_0 .

Now, since $C_0 \subset S$ is compact and $\{U_k \mid k \geq 0\}$ is an open cover of C_0 , by the compactness of C_0 , there exists a finite subcover. That is, there are finitely many sets $U_{k_1}, U_{k_2}, \dots, U_{k_m}$ such that

$$C_0 = \bigcup_{i=1}^m U_{k_i}.$$

Let $M = \max\{k_1, k_2, \dots, k_m\}$. Since the sequence $(C_k)_{k \geq 0}$ is nested, we know that

$$U_{k_1} \subset U_{k_2} \subset \dots \subset U_{k_m} = U_M.$$

Thus, we have

$$C_0 = \bigcup_{i=1}^m U_{k_i} = U_M.$$

But by definition, $U_M = C_0 \setminus C_M$, so

$$C_M = C_0 \setminus U_M = \emptyset,$$

which contradicts the assumption that each C_k is non-empty.

Therefore, the assumption that $\bigcap_{k=0}^{\infty} C_k = \emptyset$ must be false, and we conclude that

$$\bigcap_{k=0}^{\infty} C_k \neq \emptyset.$$

□

Theorem 3.6.8 *Any closed subspace of a compact space is compact.*

Proof: Let Y be a closed subspace of the compact space X , and let U_n be an open cover of Y , since Y is closed, the collection $U_n \cup \{X \setminus Y\}$ is an open cover of X which contains a finite subcover since X is compact. Therefore, there is a collection $\{n_1, \dots, n_k\}$ such that $(U_{n_k})_{k=1}^N \cup \{X \setminus Y\}$ covers X . Hence, $(U_{n_k})_{k=1}^N$ must cover Y . Therefore, Y is compact. □

Theorem 3.6.9 *Continuous images of compact spaces are compact.*

Proof: Let $f : X \rightarrow Z$ be a continuous map, where X is compact. We aim to show that $f(X)$ is compact in Z . Let $\{V_\alpha\}_{\alpha \in A}$ be an open cover of $f(X)$. Since f is continuous, the preimages $\{f^{-1}(V_\alpha)\}_{\alpha \in A}$ are open in X . Moreover, $\{f^{-1}(V_\alpha)\}_{\alpha \in A}$ forms an open cover of X because $f(X) \subseteq \bigcup_{\alpha \in A} V_\alpha$ implies that $X = f^{-1}(f(X)) \subseteq \bigcup_{\alpha \in A} f^{-1}(V_\alpha)$. Since X is compact, there exists a finite subcover $f^{-1}(V_{\alpha_1}), f^{-1}(V_{\alpha_2}), \dots, f^{-1}(V_{\alpha_n})$ that covers X . Consequently, the corresponding collection $V_{\alpha_1}, V_{\alpha_2}, \dots, V_{\alpha_n}$ is a finite cover of $f(X)$. Therefore, $f(X)$ is compact in Z . $\square$

Example 3.6.10 The closed interval $[0, 1] \subseteq \mathbb{R}$ is compact.

Example 3.6.11 The closed unit square $[0, 1] \times [0, 1]$ as a subspace of $\mathbb{R}^2$.

Definition 3.6.12 A topological space X is a *Stone* space if it is a compact totally disconnected Hausdorff space.

Definition 3.6.13 (The Cantor Set) Let $A_0 = [0, 1]$. Construct A_1 from A_0 by deleting the middle third of the segment, the open set $(\frac{1}{3}, \frac{2}{3})$. Construct A_2 from A_1 by deleting the "middle thirds", the open set $(\frac{1}{9}, \frac{2}{9}) \cup (\frac{7}{9}, \frac{8}{9})$. More generally, we define

$$A_n = A_{n-1} \setminus \bigcup_{k=0}^{2^{n-1}-1} \left(\frac{3k+1}{3^n}, \frac{3k+2}{3^n} \right).$$

The Cantor set is the set $C = \bigcap_{n \in \mathbb{N}} A_n$ as a subspace of the unit interval $[0, 1]$.

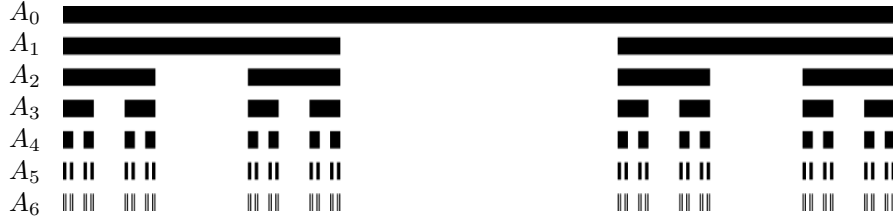


FIGURE 1. Visualization of the Cantor set

Example 3.6.14 As a compact totally disconnected Hausdorff space, the Cantor set is an example of a Stone space. As a topological space, the Cantor set is naturally homeomorphic to the product of countably many copies of the space $\{0, 1\}$, where each copy carries the discrete topology. A *Cantor Space* is the topological abstraction of the Cantor Set. A topological space is a Cantor space if it is homeomorphic to the Cantor set.

Theorem 3.6.15 Fix two real numbers a, b such that $a \leq b$. Then the interval $[a, b]$ is compact as a subspace of $\mathbb{R}$.

Proof: Assume that $\{U_\alpha\}_{\alpha \in A}$ is an open cover of $[a, b]$, meaning that each U_α is open in $\mathbb{R}$ and $[a, b] \subset \bigcup_{\alpha \in A} U_\alpha$.

Define the set

$S = \{x \in [a, b] \mid [a, x] \text{ covered by a finite number of open sets } U_\alpha \text{ from the cover}\}.$

Since $a \in [a, a]$ and the point a can be trivially covered by any open set in the cover that contains a , we have $a \in S$. Thus, S is non-empty.

Now, let $c = \sup S$, the least upper bound of S . We claim that $c = b$. Suppose, for contradiction, that $c < b$. Since $\{U_\alpha\}_{\alpha \in A}$ is an open cover of $[a, b]$, there exists an open set U_γ containing c . Because U_γ is open, there exists some $\epsilon > 0$ such that the open interval $(c - \epsilon, c + \epsilon) \subset U_\gamma$. By the definition of c as the supremum of S , for any $\delta > 0$, there exists $x \in S$ such that $c - \delta < x < c$. In particular, taking $\delta = \epsilon$, there exists $x \in S$ with $c - \epsilon < x < c$. This means that $[a, x]$ can be covered by a finite number of open sets from the cover, say $U_{\alpha_1}, U_{\alpha_2}, \dots, U_{\alpha_n}$.

Consider now the interval $[a, c + \epsilon/2]$. This interval is contained within $[a, x] \cup (c - \epsilon, c + \epsilon)$, so it can be covered by the finite collection of open sets $U_{\alpha_1}, U_{\alpha_2}, \dots, U_{\alpha_n}, U_\gamma$, where U_γ covers $(c - \epsilon, c + \epsilon)$ and therefore contains $c + \epsilon/2$. This implies that $c + \epsilon/2 \in S$, contradicting our assumption that c was the supremum of S , since we have now found a point in S greater than c . Thus, we must have $c = b$, meaning that $[a, b] = [a, c]$ can be covered by a finite number of open sets from $\{U_\alpha\}_{\alpha \in A}$. $\square$

Theorem 3.6.16 (Tube Lemma) *Let X be a compact topological space, Y be a topological space, and $X \times Y$ be their product space with the product topology. If $K \subseteq X \times Y$ is a closed subset and there exists a point $x_0 \in X$ such that $\{x_0\} \times Y \subseteq K$, then there exists an open set $U \subseteq X$ containing x_0 and an open set $V \subseteq Y$ such that $U \times V \subseteq K$.*

Proof: Let $K \subseteq X \times Y$ be a closed subset, and suppose $\{x_0\} \times Y \subseteq K$. Since K is closed, its complement $X \times Y \setminus K$ is open in the product topology on $X \times Y$. By the definition of the product topology, for each point $(x, y) \in X \times Y \setminus K$, there exist open sets $U_{x,y} \subseteq X$ and $V_{x,y} \subseteq Y$ such that $(x, y) \in U_{x,y} \times V_{x,y} \subseteq X \times Y \setminus K$.

Now consider $\{x_0\} \times Y \subseteq K$. For every $y \in Y$, the point (x_0, y) cannot belong to $X \times Y \setminus K$ because $\{x_0\} \times Y \subseteq K$. This ensures that around x_0 , no open neighborhood intersects $X \times Y \setminus K$ entirely within X . By the compactness of X , we can cover $\{x_0\}$ with finitely many such neighborhoods to construct an open set $U \subseteq X$ containing x_0 . Similarly, we can construct an open set $V \subseteq Y$ such that the product $U \times V \subseteq K$. Thus, $U \times V$ provides the desired open neighborhood in $X \times Y$ contained entirely within K , completing the proof. $\square$

Theorem 3.6.17 *Let X and Y be compact. Then $X \times Y$ is compact.*

Proof: Let X and Y be compact topological spaces. To show that $X \times Y$ is compact, consider an open cover $\{U_\alpha\}_{\alpha \in A}$ of $X \times Y$. For each $x \in X$, the set $\{x\} \times Y$ is homeomorphic to Y and therefore compact. Thus, $\{x\} \times Y$ can be covered by finitely many elements of $\{U_\alpha\}$. By the Tube Lemma, for each $x \in X$, there exists an open neighborhood V_x of x in X and finitely many open sets $U_{\alpha_1(x)}, \dots, U_{\alpha_{k(x)}(x)}$ in the cover such that

$$V_x \times Y \subseteq \bigcup_{i=1}^{k(x)} U_{\alpha_i}.$$

The collection $\{V_x\}_{x \in X}$ is an open cover of X , and since X is compact, there exists a finite subcover $V_{x_1}, V_{x_2}, \dots, V_{x_m}$ of X . For each x_i , there is a corresponding finite collection of open sets covering $V_{x_i} \times Y$, and thus $X \times Y$ is covered by finitely many sets from $\{U_\alpha\}$. This yields a finite subcover of $X \times Y$, proving that $X \times Y$ is compact. $\square$

Definition 3.6.18 A topological space X is called *extremally disconnected* if the closure of every open subset of X is open.

Definition 3.6.19 A topological space X is said to be a *separable* if it has a countable dense subset in X . In other words, is said to be separable if there is a subset A of X such that A is countable and $\bar{A} = X$.

Example 3.6.20 Consider the set of rational numbers $\mathbb{Q}$, which is a subset of $\mathbb{R}$ equipped with the standard topology. The closure of $\mathbb{Q}$ in $\mathbb{R}$ is $\mathbb{R}$ itself, because every real number can be approximated arbitrarily closely by rational numbers. This demonstrates that $\bar{\mathbb{Q}} = \mathbb{R}$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, the space $\mathbb{R}$ is separable, meaning it contains a countable dense subset (specifically, $\mathbb{Q}$).

Theorem 3.6.21 (Heine-Borel Theorem) *A set $K \subset \mathbb{R}$ is compact if and only if K is closed and bounded.*

Proof: First, assume K is compact. Compactness implies every open cover of K has a finite subcover. To show K is bounded, assume for contradiction that K is unbounded. Consider the open cover $\{(-n, n) : n \in \mathbb{N}\}$. Since K is unbounded, no finite subcollection of these intervals can cover K , contradicting compactness. Thus, K is bounded. To show K is closed, assume K is not closed, so there exists a limit point $a \notin K$. Construct an open cover $\{V_n = \{x : |x - a| > 1/n\}\}$ for K . Every point of K lies in some V_n , but no finite subcover exists since points of K can approach a . This contradiction shows K is closed.

Conversely, if K is closed and bounded, it lies within a closed, bounded interval $[a, b]$, which is compact. Since K is a closed subset of a compact set, it is compact. Thus, K is compact if and only if it is closed and bounded. $\square$

Lemma 3.6.22 *Let X be a set. Let $\mathcal{A}$ be a set ("collection") of subsets of X having the finite intersection property. Then there is a collection $\mathcal{D}$ of subsets of X such that $\mathcal{D}$ contains $\mathcal{A}$ and $\mathcal{D}$ has the finite intersection property, and no collection of subsets of X that properly contains $\mathcal{D}$ has this property. Such a collection $\mathcal{D}$ is said to be maximal with respect to the finite intersection property.*

Proof: To prove this lemma, we aim to demonstrate that for a collection $\mathcal{A}$ of subsets of a set X with the finite intersection property (FIP), there exists a collection $\mathcal{D}$ of subsets of X that is maximal with respect to the FIP. In other words, $\mathcal{D}$ contains $\mathcal{A}$, satisfies the FIP, and no proper extension of $\mathcal{D}$ preserves this property.

We start by considering the set $\mathcal{S}$ of all collections of subsets of X that contain $\mathcal{A}$ and have the FIP. We partially order $\mathcal{S}$ by set inclusion, so for any two elements $\mathcal{B}, \mathcal{C} \in \mathcal{S}$, we say $\mathcal{B} \leq \mathcal{C}$ if $\mathcal{B} \subseteq \mathcal{C}$. To find a maximal element in $\mathcal{S}$, we apply Zorn's Lemma, which requires us to show that every chain in $\mathcal{S}$ has an upper bound within $\mathcal{S}$.

Let $\{\mathcal{B}_\alpha\}_{\alpha \in I}$ be a chain in $\mathcal{S}$, where each $\mathcal{B}_\alpha$ is a collection containing $\mathcal{A}$ and has the FIP. Define $\mathcal{B} = \bigcup_{\alpha \in I} \mathcal{B}_\alpha$, the union of all collections in the chain. To confirm that $\mathcal{B}$ belongs to $\mathcal{S}$, we must check that $\mathcal{B}$ has the FIP. For any finite selection $B_1, B_2, \dots, B_n$ of sets in $\mathcal{B}$, each B_i belongs to some $\mathcal{B}_{\alpha_i}$ within the chain. Since the $\mathcal{B}_\alpha$ form a chain by inclusion, there exists an index β such that all B_i belong to $\mathcal{B}_\beta$. Because $\mathcal{B}_\beta$ has the FIP, we conclude that $B_1 \cap B_2 \cap \dots \cap B_n \neq \emptyset$. Therefore, $\mathcal{B}$ has the FIP, and $\mathcal{B} \in \mathcal{S}$.

By Zorn's Lemma, there exists a maximal element $\mathcal{D}$ in $\mathcal{S}$, which is a collection of subsets of X containing $\mathcal{A}$, satisfying the FIP, and being maximal with respect to the FIP. Thus, any proper extension of $\mathcal{D}$ would violate the FIP. This concludes the proof, establishing that $\mathcal{D}$ is indeed maximal with respect to the finite intersection property. $\square$

Lemma 3.6.23 *Let X be a set. Let D be a collection of subsets of X that is maximal with respect to the finite intersection property. Then: (a) Any finite intersection of elements of D is an element of D . (b) If A is a subset of X that intersects every element of D , then A is an element of D .*

Proof: Let X be a set, and let D be a collection of subsets of X that is maximal with respect to the finite intersection property (FIP). We aim to prove two statements.

First, we show that any finite intersection of elements of D is an element of D . Since D is maximal with respect to the finite intersection property, it follows that every finite intersection of elements in D is non-empty, and if any subset $B \subset X$ were added to D that would violate the finite intersection property, it could not be in D due to maximality. Now, take any finite collection of subsets $A_1, A_2, \dots, A_n \in D$. By the finite intersection property, the intersection $A_1 \cap A_2 \cap \dots \cap A_n \neq \emptyset$. Furthermore, since $A_1 \cap A_2 \cap \dots \cap A_n$ intersects every element in D non-trivially, it follows from the maximality of D that $A_1 \cap A_2 \cap \dots \cap A_n \in D$. This proves the first part.

Next, we show that if $A \subset X$ intersects every element of D , then $A \in D$. Suppose that $A \subset X$ and that $A \cap B \neq \emptyset$ for every $B \in D$. To demonstrate that $A \in D$, assume for contradiction that $A \notin D$. Since D is maximal with respect to the finite intersection property, adding A to D would violate the FIP. This implies there must exist some finite collection $B_1, B_2, \dots, B_n \in D$ such that $A \cap B_1 \cap B_2 \cap \dots \cap B_n = \emptyset$. However, this contradicts the assumption that A intersects every element of D , since $B_1 \cap B_2 \cap \dots \cap B_n \in D$ by part (a), and thus A would need to intersect $B_1 \cap B_2 \cap \dots \cap B_n$ non-trivially. This contradiction implies that $A \in D$. $\square$

Theorem 3.6.24 (Tychonoff Theorem) *Let $X = \prod_{\alpha \in J} X_\alpha$ be the product of compact topological spaces with the product topology. Then the topological space X is compact.*

Proof: Suppose $\{U_i\}_{i \in I}$ is an open cover of X that has no finite subcover. We will aim for a contradiction to show that such an assumption leads to an impossibility. For each finite subset $I_0 \subseteq I$, define the closed set $C_{I_0} = X \setminus \bigcup_{i \in I_0} U_i$. Since $\{U_i\}_{i \in I}$ is an open cover without a finite subcover, each C_{I_0} is nonempty.

The collection $\{C_{I_0}\}$ has the *finite intersection property* (FIP) because the intersection of finitely many such sets is nonempty:

$$\bigcap_{k=1}^n C_{I_k} = X \setminus \bigcup_{k=1}^n \bigcup_{i \in I_k} U_i = X \setminus \bigcup_{i \in I'} U_i,$$

where $I' = \bigcup_{k=1}^n I_k$. Since I' is finite and $\{U_i\}$ subcover, this intersection is nonempty.

Let S be the set of all finite subsets of J . For each $F \in S$, consider the projection map

$$\pi_F : X \rightarrow X_F = \prod_{\alpha \in F} X_\alpha.$$

Since F is finite and each X_α is compact, X_F is compact. For each C_{I_0} , the image $\pi_F(C_{I_0})$ is a closed subset of X_F . Moreover, the collection $\{\pi_F(C_{I_0})\}$ retains the finite intersection property because projections preserve intersections:

$$\bigcap_{k=1}^n \pi_F(C_{I_k}) = \pi_F \left(\bigcap_{k=1}^n C_{I_k} \right) \neq \emptyset.$$

Since X_F is compact and the $\pi_F(C_{I_0})$ have the finite intersection property, the intersection over all I_0 is nonempty:

$$\bigcap_{I_0} \pi_F(C_{I_0}) \neq \emptyset.$$

Therefore, for each finite $F \subseteq J$, there exists a point $x_F \in X_F$ such that

$$x_F \in \bigcap_{I_0} \pi_F(C_{I_0}).$$

For finite subsets $F \subseteq F' \subseteq J$, the projections are compatible:

$$\pi_{F,F'} : X_{F'} \rightarrow X_F,$$

and satisfy $\pi_F = \pi_{F,F'} \circ \pi_{F'}$. From the way x_F and $x_{F'}$ are chosen, it follows that

$$\pi_{F,F'}(x_{F'}) = x_F.$$

This compatibility allows us to define a point $x \in X$ such that

$$\pi_F(x) = x_F \quad \text{for all finite } F \subseteq J.$$

Since $x \in X$, and $x_F \in \pi_F(C_{I_0})$ for all finite F and I_0 , it follows that

$$x \in \bigcap_{I_0} C_{I_0}.$$

But this implies that $x \notin U_i$ for any $i \in I$, contradicting the assumption that $\{U_i\}$ is an open cover of X . The assumption that $\{U_i\}$ has no finite subcover leads to a contradiction. Therefore, every open cover of X must have a finite subcover, and thus X is compact. $\square$

Theorem 3.6.25 *Every compact Hausdorff space is normal.*

Proof: Let X be a compact Hausdorff space. We need to show that X is normal, i.e., if A and B are two disjoint closed subsets of X , then there exist disjoint open sets U and V such that $A \subseteq U$ and $B \subseteq V$.

Since X is Hausdorff, for any pair of distinct points $x \in A$ and $y \in B$, there exist disjoint open sets U_x and V_y such that $x \in U_x$ and $y \in V_y$.

For each $x \in A$, there exists an open set U_x such that $x \in U_x$ and $U_x \cap B = \emptyset$ (since B is closed and disjoint from A). Similarly, for each $y \in B$, there exists an open set V_y such that $y \in V_y$ and $V_y \cap A = \emptyset$.

The family $\{U_x : x \in A\}$ is an open cover of A , and the family $\{V_y : y \in B\}$ is an open cover of B . Since A and B are closed in the compact space X , they are also compact. Hence, there exist finite subcovers for A and B . That is, we can select finite sets $\{U_{x_1}, U_{x_2}, \dots, U_{x_n}\}$ that cover A and $\{V_{y_1}, V_{y_2}, \dots, V_{y_m}\}$ that cover B .

Define the open set $U = \bigcup_{i=1}^n U_{x_i}$ to cover A , and $V = \bigcup_{j=1}^m V_{y_j}$ to cover B . By construction, $A \subseteq U$, $B \subseteq V$, and $U \cap B = \emptyset$, $V \cap A = \emptyset$. Moreover, U and V are open sets, since they are finite unions of open sets, and they are disjoint by construction. Thus, every compact Hausdorff space is normal. $\square$

4 FUNDAMENTAL COMPACTIFICATIONS

Remark: The process of compactification is central to extending spaces in a meaningful way to ensure certain desirable properties, such as compactness, while preserving the original structure of the space. The motivation for studying compactifications arises from the desire to take non-compact spaces and embed them into compact ones without losing the essence of their topology. This process allows for the extension of various theorems, such as those concerning continuous functions, to spaces where compactness does not originally hold. The idea of compactification is particularly valuable in areas where completeness or finiteness is required for the applicability of certain results, such as in the study of function spaces, boundary behavior in partial differential equations, and cosmological models in mathematical physics.

Various types of compactifications, such as the one-point compactification, Stone-Čech compactification, and N -point compactifications, each provide different insights and tools depending on the structure of the underlying space. These constructions not only broaden the theoretical framework of topology but also find applications in other fields like algebra, set theory, and even Boolean algebras.

Definition 4.0.1 A *compactification* of a topological space X is a compact Hausdorff space Y containing X as a subspace such that $\bar{X} = Y$, that is, $X \subseteq Y$ is a dense subspace. Two compactifications Y_1 and Y_2 of X are said to be *equivalent* if there is a homeomorphism $h : Y_1 \rightarrow Y_2$ such that $h(x) = x$ for every $x \in X$.

4.1 One-Point Compactification.

Remark: One of the simplest and most illustrative examples of compactification is the one-point compactification, also called the *Alexandroff extension*,² which transforms a given space into a compact space by the addition of a single, idealized point at "infinity." The one-point compactification is essential for studying infinite spaces in a manageable way. By adding just one point, we gain a tool that simplifies the study of properties like continuity, convergence, and boundary conditions. In many applications, this compactification method allows mathematicians to bridge the gap between infinite and finite spaces, providing a way to analyze global properties without being overwhelmed by the unbounded nature of the original space. The one-point compactification forms the basis for more complex types of compactifications and serves as an intuitive example of how topology can extend familiar spaces into richer mathematical frameworks.

In the case of the real line, we can imagine bending the number line into a circle and adding one point to "close" it at infinity. This added point creates a seamless compact space where the line no longer stretches indefinitely but instead loops back on itself in a circular fashion. This process demonstrates how the real line, while infinite in extent, can be compactified by incorporating this idealized point, turning the open structure of the number line into a closed, finite form.

²Named after the Russian mathematician Pavel Alexandroff (1896 - 1982).

Definition 4.1.1 Let X be a topological space that is *locally compact* and *Hausdorff* but *not compact*. The *one-point compactification* of X , denoted X^* , is defined as:

$$X^* = X \cup \{\infty\}$$

where ∞ is a new point not in X , often referred to as the *point at infinity*.

The topology on X^* is defined by the following conditions:

- (1) Every open set $U \subseteq X$ in the original topology of X remains open in X^* .
- (2) A set $U \subseteq X^*$ containing the point ∞ is open if and only if $X^* \setminus U$ is compact in X .

With this topology, X^* becomes a compact Hausdorff space, and the subset X is an open dense subset of X^* . This point ∞ serves as a point "at infinity," thereby compactifying X by effectively "closing it off" with this additional point.

A space X has a compactification if and only if X is completely regular.

Theorem 4.1.2 *The Alexandroff extension is a topology on X^* .*

Proof: To prove that the Alexandroff extension is a topology on $X^* = X \cup \{\infty\}$, we need to verify that it satisfies the axioms of a topology. Let X be a non-compact topological space with topology τ . The *Alexandroff extension* (or 1-point compactification) of X , denoted $X^* = X \cup \{\infty\}$, is constructed by adding a single point ∞ to X . The topology τ^* on X^* is defined as follows:

- (1) Every open set $U \in \tau$ is also an open set in τ^* .
- (2) For any closed set $C \subseteq X$ such that $X \setminus C$ is compact, the set $X^* \setminus C = (X \setminus C) \cup \{\infty\}$ is open in τ^* .

Thus, the open sets in τ^* include all open sets of X along with sets of the form $U \cup \{\infty\}$, where $U \subset X$ is open in X and $X \setminus U$ is compact.

We verify the topology axioms as follows. First, we show that the empty set and X^* are in τ^* . Since $\emptyset \in \tau$, it follows that $\emptyset \in \tau^*$. The set $X^* = X \cup \{\infty\}$ is also open in τ^* because we can write $X^* = X \cup \{\infty\}$, where $X \in \tau$ and $X \setminus X = \emptyset$ is compact, implying $X^* \in \tau^*$. Therefore, both the empty set and X^* are elements of τ^* .

Next, we check that τ^* is closed under finite intersections. Let $A, B \in \tau^*$. There are two cases to consider. First, if $A, B \in \tau$ (i.e., A and B are open subsets of X), then $A \cap B \in \tau$ because τ is a topology on X , so $A \cap B \in \tau^*$. For the second case, assume $A = U \cup \{\infty\}$ and $B = V \cup \{\infty\}$, where $U, V \subset X$ are open in X and $X \setminus U$ and $X \setminus V$ are compact. Then

$$A \cap B = (U \cap V) \cup \{\infty\}.$$

Since $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$, which is a finite union of compact sets and therefore compact, we conclude that $(U \cap V) \cup \{\infty\} \in \tau^*$. Thus, τ^* is closed under finite intersections.

Finally, we verify that τ^* is closed under arbitrary unions. Let $\{A_i\}_{i \in I}$ be a collection of open sets in τ^* . We consider two cases. First, if $\infty \notin A_i$ for all $i \in I$, then $A_i \in \tau$ for each $i \in I$, so $\bigcup_{i \in I} A_i \in \tau \subset \tau^*$. In the second case, suppose there exists some $j \in I$ such that $\infty \in A_j$. Then $A_j = U_j \cup \{\infty\}$ for some open set $U_j \subset X$ where $X \setminus U_j$ is compact. We can write

$$\bigcup_{i \in I} A_i = \left(\bigcup_{i \in I} (A_i \cap X) \right) \cup \{\infty\}.$$

Since $\bigcup_{i \in I} (A_i \cap X) \subset X$ is open in X and contains U_j , its complement in X is compact (as it is contained within the compact set $X \setminus U_j$). Therefore, $\bigcup_{i \in I} A_i \in \tau^*$, proving that τ^* is closed under arbitrary unions.

Since τ^* satisfies all three conditions for a topology, we conclude that τ^* is indeed a topology on X^* . This completes the proof. $\square$

Theorem 4.1.3 *The Alexandroff extension is compact.*

Proof: Let $X = (X, \tau)$ be a topological space. Let $X^* = X \cup \{p\}$ and p be a new element not in X . Let $X^* = (S^*, \tau^*)$. Let $\mathcal{U}$ be an open cover of X^* . Then at least one $V \in \mathcal{U}$ contains p . Since $p \notin S$ we know V is not an open set of X . Thus, by the definition of Alexandroff extension, V must be the complement relative to S^* of a closed, compact subset of the form $V = (X \setminus S^*) \cup \{p\}$. Then since the complement V is compact, it can be covered by a finite number of sets of $\mathcal{U}$. So X^* can be covered by V and that a finite number of sets in $\mathcal{U}$ are a cover for the complement V . That is, $\mathcal{U}$ has a finite subcover. Therefore, the result by definition is a compact topological space. $\square$

Theorem 4.1.4 *If X is a Hausdorff and locally compact space which is not compact, then the Alexandroff extension $Y = X \cup \{\infty\}$ is the unique one-point compactification of X which is Hausdorff.*

Proof: Let X be a Hausdorff, locally compact space that is not compact. The Alexandroff extension $Y = X \cup \{\infty\}$ is defined by adding a single point ∞ to X and specifying that a subset $U \subset Y$ is open in Y if and only if U is open in X or U contains ∞ and $Y \setminus U$ is compact in X .

First, we show that Y is compact. Let $\{U_\alpha\}$ be an open cover of Y . Since X is not compact, the point $\infty \in Y$ cannot be covered by open sets from X alone, so some open set U_{α_0} in the cover must contain ∞ . The complement $Y \setminus U_{\alpha_0}$ is compact in X , and the collection $\{U_\alpha : \alpha \neq \alpha_0\}$ covers this compact set. Thus, there exists a finite subcover $\{U_{\alpha_1}, \dots, U_{\alpha_n}\}$ covering $Y \setminus U_{\alpha_0}$. Together with U_{α_0} , we obtain a finite subcover of Y , proving that Y is compact.

Next, we show that Y is Hausdorff. Take any two distinct points $x, y \in Y$. If both x and y are in X , then since X is Hausdorff, there exist disjoint open sets $U, V \subset X$ such that $x \in U$ and $y \in V$, which remain open in Y and separate x and y . If $x = \infty$ and $y \in X$, then since X is locally compact, there exists an open neighborhood $W \subset X$ of y such that $\overline{W}$ is compact in X . The set $Y \setminus \overline{W}$ is open in Y and contains ∞ , while W is open in Y and contains y . These disjoint open sets separate ∞ and y , so Y is Hausdorff.

To show uniqueness, let Z be any compact Hausdorff space containing X as an open subspace such that $Z \setminus X$ consists of a single point, say p . Define a map $f : Y \rightarrow Z$ by setting $f(x) = x$ for $x \in X$ and $f(\infty) = p$. This map is well-defined and bijective, as $Y \setminus X = \{\infty\}$ and $Z \setminus X = \{p\}$. To show f is a homeomorphism, we verify continuity of f and f^{-1} . If $V \subset Z$ is open and does not contain p , then $V \subset X$ and is open in Y by construction. If V contains p , then $Z \setminus V$ is compact in X , and $f^{-1}(V) = Y \setminus (Z \setminus V)$ is open in Y , showing f is continuous. Similarly, for $U \subset Y$, if $\infty \notin U$, then $U \subset X$ and is open in Z . If $\infty \in U$, then $Y \setminus U$ is compact in X , and $f(U) = Z \setminus (Y \setminus U)$ is open in Z , proving f^{-1} is continuous. Thus, f is a bijective homeomorphism, and Y and Z are homeomorphic.

This establishes that $Y = X \cup \{\infty\}$ is the unique one-point compactification of X which is Hausdorff. $\square$

Example 4.1.5 The one-point compactification of the set of positive integers is homeomorphic to the space consisting of $K = \{0\} \cup \{\frac{1}{n} | n \in \mathbb{Z}^+\}$ with the order topology.

Example 4.1.6 The one-point compactification of $\mathbb{R}^2$ is homeomorphic to the sphere S^2 .

Example 4.1.7 The one-point compactification of $\mathbb{R}^1$ is homeomorphic to the circle S^1 .

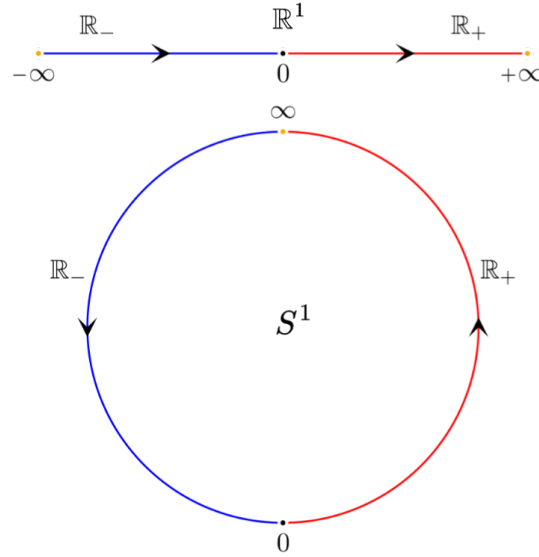


FIGURE 2. One-Point Compactification (Alexandroff Extension) of $\mathbb{R}^1$

4.2 N -Point Compactification. We can continue this extrapolation and generalize it to any number n .

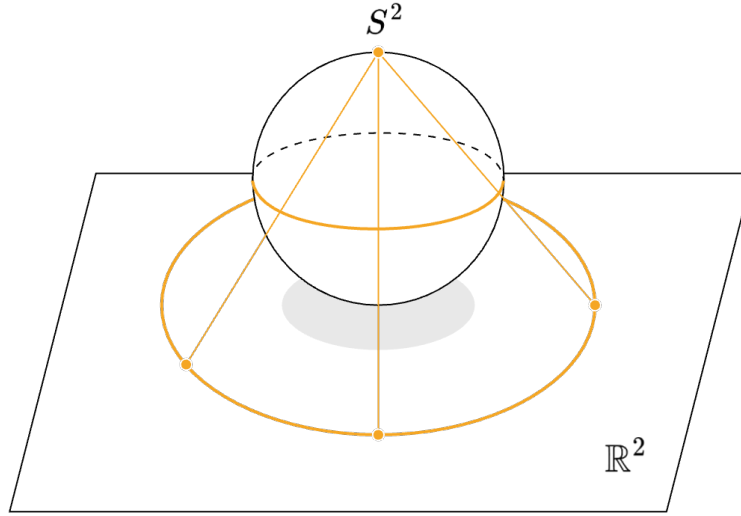
Definition 4.2.1 [8] Let X be a non-compact, locally compact Hausdorff space, and let $N \in \mathbb{N}$ be a positive integer. The N -point compactification of X , denoted by X_N , is a compact Hausdorff space obtained by adding N distinct points at infinity to X in the following manner:

- (1) *Set Construction:* Define

$$X_N = X \cup \{p_1, p_2, \dots, p_N\},$$

where $\{p_1, p_2, \dots, p_N\}$ is a set of N distinct points not contained in X , often referred to as "points at infinity."

- (2) *Topology on X_N :* The topology on X_N is defined by specifying open sets as follows:

FIGURE 3. One-Point Compactification (Alexandroff Extension) of $\mathbb{R}^2$

- Every open set $U \subset X$ in the original topology of X is also an open set in X_N .
- For each $p_i \in \{p_1, p_2, \dots, p_N\}$, a neighborhood of p_i is defined as

$$U_{p_i} = X_N \setminus K,$$

where $K \subset X$ is a compact subset. The collection of all such sets U_{p_i} forms a neighborhood basis for p_i .

- (3) *Compactness:* With this topology, X_N is a compact Hausdorff space.
- (4) *Universal Property:* The N -point compactification X_N has the following universal property: Let Y be any compact Hausdorff space, and let $f : X \rightarrow Y$ be a continuous map. Then f uniquely extends to a continuous map $\bar{f} : X_N \rightarrow Y$ if and only if $f(X)$ is dense in Y .

This construction generalizes the notion of the one-point compactification by adding N distinct points, resulting in a compact space that contains X as a dense subset.

Theorem 4.2.2 *An infinite discrete space has infinitely many N -point compactifications for each positive integer $N > 1$.*

Proof: Let X be an infinite discrete space. Since X is discrete, every subset of X is open, and the only compact subsets of X are finite subsets. For an N -point compactification X_N of X , we add N points $\{p_1, p_2, \dots, p_N\}$, called "points at infinity," to X . We define the topology on $X_N = X \cup \{p_1, p_2, \dots, p_N\}$ such that each open set in X remains open in X_N , and for each p_i , a neighborhood of p_i in X_N is defined as $U_{p_i} = X_N \setminus K$, where $K \subset X$ is a finite subset. Thus, neighborhoods of p_i exclude different finite subsets of X .

To construct distinct N -point compactifications, we will associate the "points at infinity" $\{p_1, p_2, \dots, p_N\}$ with distinct infinite subsets of X . Let $\mathcal{A} = \{A_\alpha : \alpha \in \mathbb{N}\}$ be a collection of disjoint infinite subsets of X such that $X = \bigcup_{\alpha \in \mathbb{N}} A_\alpha$. (For example, if X is countably infinite, we can partition X into countably infinitely

many disjoint infinite subsets.) For each subset $A_\alpha \in \mathcal{A}$, construct a distinct N -point compactification X_{N,A_α} of X by associating the neighborhoods of the points $\{p_1, p_2, \dots, p_N\}$ with the finite subsets of A_α . Specifically, define the neighborhoods of each p_i in X_{N,A_α} as $U_{p_i} = X_{N,A_\alpha} \setminus K$, where $K \subset A_\alpha$ is a finite subset. In other words, in the compactification X_{N,A_α} , the points at infinity $\{p_1, \dots, p_N\}$ are "associated" with the infinite subset A_α , and their neighborhoods avoid finite subsets of A_α .

To show that these compactifications X_{N,A_α} are distinct, consider any two different compactifications X_{N,A_α} and X_{N,A_β} with $A_\alpha \neq A_\beta$. Since A_α and A_β are disjoint, there exists no homeomorphism between X_{N,A_α} and X_{N,A_β} that preserves the topology of the points at infinity. In particular, the neighborhoods of p_i in X_{N,A_α} are defined by complements of finite subsets of A_α , whereas the neighborhoods of p_i in X_{N,A_β} are defined by complements of finite subsets of A_β . Therefore, each compactification X_{N,A_α} is topologically distinct from X_{N,A_β} for $\alpha \neq \beta$.

Since we can choose infinitely many distinct subsets A_α (from the partition $\mathcal{A}$), there are infinitely many distinct choices for the N -point compactification X_{N,A_α} . Therefore, for each integer $N > 1$, the infinite discrete space X has infinitely many N -point compactifications, as required. $\square$

5 STONE-ČECH COMPACTIFICATION

Remark: The compactification constructed by Čech and Stone's called the Stone-Čech compactification; and is written as βX . A natural question that may arise from the knowledge of one point compactification being the smallest compactification, what is the largest compactification you can take?

5.1 Construction via the Unit Interval.

Definition 5.1.1 A *cube* is a space $I^\kappa = [0, 1]^\kappa$, where κ is some cardinal number. A subspace of a cube is any subset of such space with the subspace topology.

Theorem 5.1.2 For any given space X , it is regular if and only if there exists a cube $[0, 1]^\kappa$ for some index set κ such that X is homeomorphic to a subspace of $[0, 1]^\kappa$. In other words, a space X is regular if and only if it is homeomorphic to a subspace of a cube.

Proof: If X is completely regular then by definition one point sets are closed and there is a family of continuous functions each mapping X to $[0, 1]$ which separate points from closed sets. By the Embedding Theorem, there is an embedding of X in $[0, 1]^\kappa$. If X is homeomorphic to a subspace of $[0, 1]^\kappa$, since $[0, 1]^\kappa$ is a metric space, it is Hausdorff and so by Theorem 2.60, each one-point set is closed. By Lemma 3.5.10, $\mathbb{R}^\kappa$ under the box topology is completely regular. Since the box topology is finer than the product topology, each $f : X \mapsto [0, 1]$ in the definition of completely regular which is continuous in the box topology is also continuous in the product topology. Therefore, $\mathbb{R}^\kappa$ under the product topology is completely regular and hence X is completely regular. $\square$

Lemma 5.1.3 Let X be a completely regular space. Given any continuous map $f : X \rightarrow [a, b]$, we can extend the map uniquely to a continuous map $g : \beta X \rightarrow [a, b]$.

Proof: By scaling and shifting, we may WLOG consider $[a, b] = [0, 1]$. Then our continuous function f belongs to the collection $\{f_\alpha\}_{\alpha \in J}$ described in previous theorem; suppose $f = f_\beta$ for some $\beta \in J$. Then the composed map

$$\beta X \hookrightarrow [0, 1]^J \xrightarrow{\pi_\beta} [0, 1]$$

where π_β is the projection map, is our desired continuous extension g . $\square$

Lemma 5.1.4 *Let $A \subseteq X$ and let $f : A \rightarrow Z$ be a continuous map of A into a Hausdorff space Z . Then there is at most one extension of f to a continuous function $g : \bar{A} \rightarrow Z$.*

Proof: Suppose $g, g' : \bar{A} \rightarrow Z$ are two continuous extensions of f with some $x \in \bar{A}$ satisfying $g(x) \neq g'(x)$. Since Z is Hausdorff, pick U and U' disjoint neighborhoods of $g(x)$ and $g'(x)$, respectively. By the continuity of f , there exist V and V' neighborhoods of x such that $g(V) \subseteq U$ and $g'(V') \subseteq U$. Being a neighborhood of $x \in \bar{A}$, $V \cap V'$ contains an element of A , say a . Then $g(a) \in g(V) \subseteq U$ and $g'(a) \in g'(V') \subseteq U'$ but $g(a) = f(a) = g'(a)$, a contradiction. $\square$

The Stone-Čech Compactification satisfies the following universal property which explains its central importance.

Theorem 5.1.5 *Let X be a completely regular space. Given any continuous map $f : X \rightarrow C$ of X to a compact Hausdorff space C , the map f extends uniquely to a continuous map $g : \beta X \rightarrow C$.*

Proof: If C is compact Hausdorff, then C is also completely regular and can be embedded into some cube $[0, 1]^K$. For each $\gamma \in K$, define f_γ as the composed map

$$X \xrightarrow{f} C \hookrightarrow [0, 1]^K \xrightarrow{\pi_\gamma} [0, 1]$$

By lemma 5.1.3, each f_γ can be extended to a continuous map $g_\gamma : \beta X \rightarrow [0, 1]$. By defining $g = (g_\gamma)_{\gamma \in K}$ we obtain a continuous map $g : \beta X \rightarrow [0, 1]^K$ that is an extension of $X \xrightarrow{f} C \hookrightarrow [0, 1]^K$. To finish, we need to check that $g(\beta X) \subseteq C$. Indeed, by continuity of g (Lemma 2.24),

$$g(\beta X) = g(\bar{X}) \subseteq \overline{g(X)} = \overline{f(X)} \subseteq \bar{C} = C$$

.

$\square$

Definition 5.1.6 Let X be completely regular space. Let βX be a compactification of X such that for any compact Hausdorff space K and any continuous mapping $f : X \rightarrow K$ we have a continuous map $\beta f : \beta X \rightarrow K$ that acts as a unique extension of f . We call βX the Stone-Čech Compactification.

5.2 Construction via Ultrafilters.

Definition 5.2.1 A *filter* $\mathcal{F}$ on a set S is a subset of the power set $\mathcal{P}(S)$ such that

- (1) $\emptyset \notin \mathcal{F}$ and $S \in \mathcal{F}$
- (2) $A, B \in \mathcal{F}$ implies $A \cap B \in \mathcal{F}$
- (3) $A \in \mathcal{F}$ and $A \subseteq B \subseteq S$ implies $B \in \mathcal{F}$

A filter is said to be an *ultrafilter* if it is maximal among all filters.

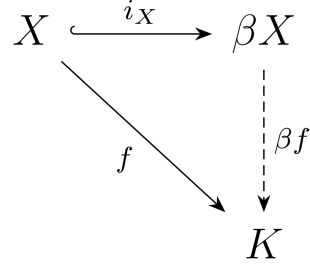


FIGURE 4. Commutative Diagram of the Universal Property of the Stone-Čech Compactification

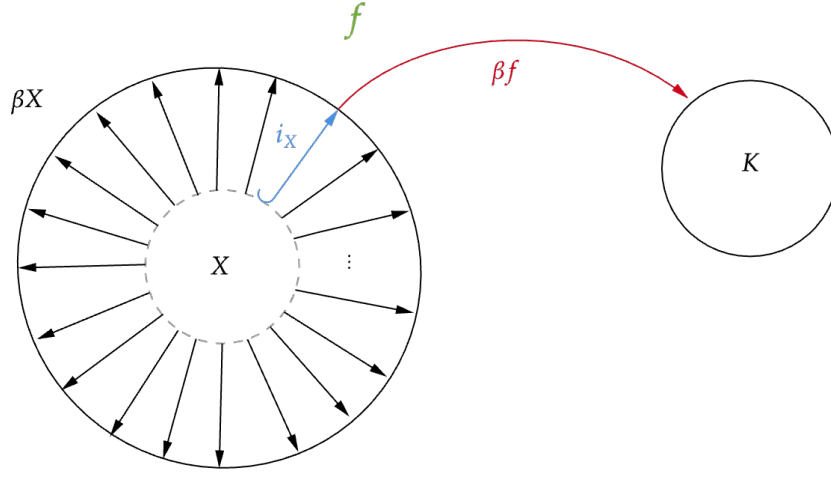


FIGURE 5. Stone-Čech Compactification (Stone-Čech Extension) of a non-compact space X .

Observation 5.1 A filter $\mathcal{F}$ on a set S has the *Finite Intersection Property (FIP)*: whenever $A_1, \dots, A_n \in \mathcal{F}$, we have $A_1 \cap \dots \cap A_n$ non-empty.

Definition 5.2.2 A filter $\mathcal{F}$ on a set S is *principal* if there exists a principal subset $M \subseteq S$ such that $\mathcal{F} = \{A \subseteq S : M \subseteq A\}$.

Definition 5.2.3 A filter $\mathcal{F}$ on a set S is said to be *free* if $\bigcap_{A \in \mathcal{F}} A = \emptyset$.

Example 5.2.4 Let S be an infinite set. The *Fréchet filter* $\mathcal{F}$ on S is defined as $\mathcal{F} = \{A \subseteq S : S \setminus A \text{ is finite}\}$.

Theorem 5.2.5 A filter $\mathcal{F}$ on an infinite set S is free if and only if it contains the Fréchet filter on S .

Proof: Clearly containing the Fréchet filter implies that the filter is free. Let us prove the other direction by proving the contrapositive. Let $\mathcal{F}$ be a filter that does not contain the Fréchet filter. Then there exists a finite set B such that $A := S \setminus B$ is not in $\mathcal{F}$. We claim that some element of B is in the intersection. Indeed, if we assume on the contrary that for every $b \in B$ there is $F_b \in \mathcal{F}$ such that $b \notin F_b$, then $\bigcap_{b \in B} F_b$ is an element of $\mathcal{F}$ that is a subset of A . This contradicts the axiom that filters are upward closed. $\square$

Example 5.2.6 Now we show that it is possible for a filter to be neither free nor principal. Let E be the set of even integers, $\mathcal{F}'$ a Fréchet filter on the set of odd integers, and define $\mathcal{F} = \{E \cup F : F \in \mathcal{F}'\}$. First, $\mathcal{F}$ is not free because $\bigcap_{A \in \mathcal{F}} A = E$. Second, $\mathcal{F}$ is not principal because what can a principal subset M be? We cannot set $M = E$ because $E \notin \mathcal{F}$ and if we add any odd integer n to M , there are elements of $\mathcal{F}$ that exclude n . However, the dichotomy of principal-versus-free holds true for ultrafilters.

Theorem 5.2.7 *An ultrafilter $\mathcal{F}$ on a set S must be either principal or free.*

Proof: Since every filter on a finite set is principal, we may take S infinite. Suppose $\mathcal{F}$ is not principal. We shall show that $\mathcal{F}$ contains the Fréchet filter. Now, given any $s \in S$, the ultrafilter $\mathcal{F}$ either contains $\{s\}$ or $S \setminus \{s\}$. Since $\mathcal{F}$ is not principal, it cannot contain $\{s\}$, so it must contain $S \setminus \{s\}$. For any finite set B , we note that $S \setminus B = \bigcap_{b \in B} (S \setminus \{b\})$. As such, $S \setminus B \in \mathcal{F}$. This completes the proof that $\mathcal{F}$ contains the Fréchet filter. $\square$

Definition 5.2.8 A filter basis on a set S is a subset $\mathcal{B}$ of the power set $\mathcal{P}(S)$ such that $\mathcal{B}$ is non-empty $\mathcal{B}$ does not contain the empty set. If $A_1, A_2 \in \mathcal{B}$, then $A_1 \cap A_2$ contains a member of $\mathcal{B}$. Given a filter basis $\mathcal{B}$, we can check that the family of subsets $\mathcal{F} := \{A \subseteq S : B \subseteq A \text{ for some } B \in \mathcal{B}\}$ forms a filter. It can also be checked that this is the minimum filter containing $\mathcal{B}$ and it is called the filter generated by $\mathcal{B}$.

Definition 5.2.9 If $\mathcal{B}, \mathcal{C}$ are filter bases on a set S , we say that $\mathcal{C}$ is *finer* than $\mathcal{B}$ if for every $B \in \mathcal{B}$, there exists $C \in \mathcal{C}$ such that $C \subseteq B$. If $\mathcal{B}$ is also finer than $\mathcal{C}$, then we say that $\mathcal{B}$ and $\mathcal{C}$ are equivalent filter bases.

Definition 5.2.10 A filter subbasis on a set S is a subset $\mathcal{A}$ of the power set $\mathcal{P}(S)$ such that $\mathcal{A}$ is non-empty. Finite intersections of elements of $\mathcal{A}$ is non-empty. Given a filter subbasis $\mathcal{A}$, we can check that the family of subsets $\mathcal{B}$ constructed by taking finite intersections of elements from $\mathcal{A}$ forms a filter basis. The filter generated by $\mathcal{B}$ is said to be the filter generated by the subbasis $\mathcal{A}$, and is the minimum filter containing $\mathcal{A}$.

Theorem 5.2.11 *Given an ultrafilter $\mathcal{F}$ on a set S and a subset $A \subseteq S$, either $A \in \mathcal{F}$ or $S \setminus A \in \mathcal{F}$ (and only exactly one).*

Proof: First, it is easy to see that $\mathcal{F}$ cannot contain A and $S \setminus A$ both together because $A \cap (S \setminus A) = \emptyset$. Now, assume on the contrary that neither is in $\mathcal{F}$. I claim that either A or $S \setminus A$ can be added to $\mathcal{F}$ to obtain a subbasis. Indeed, if A cannot be added, it means there exists $F_1, \dots, F_n \in \mathcal{F}$ such that $A \cap (F_1 \cap \dots \cap F_n) = \emptyset$, so $F := F_1 \cap \dots \cap F_n \in \mathcal{F}$ is contained in $S \setminus A$. Similarly, if $S \setminus A$ cannot be added, there exists some $F' \in \mathcal{F}$ contained in A . It is not possible to have such F and F' in $\mathcal{F}$ at the same time, because $F \cap F' = \emptyset$.

Now that it is possible to obtain a subbasis strictly larger than $\mathcal{F}$, the new subbasis generates a strictly larger filter than $\mathcal{F}$, which contradicts its maximality. $\square$

The following lemma is a weak form of the Axiom of Choice.

Lemma 5.2.12 (Ultrafilter lemma) *Every proper filter $\mathcal{F}$ is contained in an ultrafilter.*

Proof: We apply Zorn's Lemma to the poset of filters on X that contain F . Define $\mathcal{P}$ to be the set of all filters on X that contain F , ordered by inclusion. We aim to show that $\mathcal{P}$ has a maximal element, which will be an ultrafilter containing F . To apply Zorn's Lemma, we need to verify that every chain (totally ordered subset) $\mathcal{C} \subseteq \mathcal{P}$ has an upper bound in $\mathcal{P}$. Define

$$G = \bigcup_{F' \in \mathcal{C}} F'.$$

We claim that G is a filter on X and that $G \in \mathcal{P}$. To show that G is a filter, we verify the filter axioms. Since $F \subseteq G$ and $F \neq \emptyset$, G is non-empty. If $A, B \in G$, then there exist $F_A, F_B \in \mathcal{C}$ such that $A \in F_A$ and $B \in F_B$. Since $\mathcal{C}$ is a chain, assume without loss of generality that $F_A \subseteq F_B$, so $A, B \in F_B$ and $A \cap B \in F_B \subseteq G$. If $A \in G$ and $A \subseteq B \subseteq X$, there exists $F_A \in \mathcal{C}$ such that $A \in F_A$. Since F_A is a filter, $B \in F_A \subseteq G$. Thus, G is a filter containing F . By construction, $G \supseteq F'$ for every $F' \in \mathcal{C}$, so G is an upper bound of $\mathcal{C}$ in $\mathcal{P}$. By Zorn's Lemma, $\mathcal{P}$ has a maximal element, say U , which is a filter on X containing F . Now, we claim that U is an ultrafilter.

Suppose, for contradiction, that U is not an ultrafilter. Then there exists a subset $A \subseteq X$ such that $A \notin U$ and $X \setminus A \notin U$. Define U' as the filter generated by $U \cup \{A\}$, i.e., the smallest filter containing $U \cup \{A\}$. Since $X \setminus A \notin U$, $U \cup \{A\}$ satisfies the Finite Intersection Property. Hence, the filter U' generated by $U \cup \{A\}$ is proper. Since $U \subsetneq U'$, this contradicts the maximality of U in $\mathcal{P}$. Therefore, U must be an ultrafilter containing F . $\square$

Theorem 5.2.13 (Ultrafilter convergence theorem) *Let Y be a topological space.*

1. *Y is compact if and only if every ultrafilter $\mathcal{F}$ on Y converges to at least one point.*
2. *Y is Hausdorff if and only if every ultrafilter $\mathcal{F}$ on Y converges to at most one point.*

Proof: 1) Suppose for contradiction that Y is compact, but $\mathcal{F}$ has no limit points. Then for all $y \in Y$, there is an open set U_y containing y such that $U_y \notin \mathcal{F}$. So $Y = \bigcup_y U_y$, and by compactness, $Y = \bigcup_{i=1}^n U_{y_i}$. But $Y \in \mathcal{F}$, so some $U_{y_i} \in \mathcal{F}$, a contradiction. Conversely, suppose that Y is not compact. Then there is an open cover $Y = \bigcup_i U_i$ with no finite subcover. So $\bigcap_i (Y \setminus U_i) = \emptyset$, but no finite subintersection is empty. Then $(Y \setminus U_i)_i$ has the FIP, so we can take an ultrafilter $\mathcal{F}$ containing it. Now for any point $y \in Y$, y is contained in some U_i , and $U_i \notin \mathcal{F}$, since $(Y \setminus U_i) \in \mathcal{F}$. So y is not a limit point of $\mathcal{F}$.

2) Suppose for contradiction that Y is Hausdorff, but $y \neq y'$ are both limit points of $\mathcal{F}$. Take disjoint open sets $y \in U, y' \in U'$. Now $U, U' \in \mathcal{F}$, but $U \cap U' = \emptyset$, contradiction. Conversely, suppose that Y is not Hausdorff. Then there are points $y \neq y'$ such that every open neighborhood of y intersects every open neighborhood of y' . This means that $U \cup U'$ has the FIP. Let $\mathcal{F}$ be an ultrafilter containing it. Then y and y' are both limit points of $\mathcal{F}$. $\square$

So if $f : X \rightarrow Y$ is a function from a set to a compact Hausdorff space, f has a unique $\mathcal{F}$ -limit for every ultrafilter $\mathcal{F}$ on X . If $\mathcal{F}$ is the principal filter generated by $x \in X$, then the $\mathcal{F}$ -limit of f is $f(x)$.

If the space X has the discrete topology (hence also completely regular), there is another way of describing the Stone-Čech Compactification using ultrafilters (a topological space is fundamentally a set). Anticipating the final result, let βX denote the set of all ultrafilters on X . We define a topology on βX as follows.

Definition 5.2.14 For a subset $A \subseteq X$, the *Stone set* $\hat{A} \subseteq \beta X$ corresponding to A is the set of ultrafilters described by $\hat{A} = \{\mathcal{F} \in \beta X : A \in \mathcal{F}\}$. The *Stone base* is a basis for the topology on βX given by the collection of all stone sets $\mathcal{B} = \{\hat{A} : A \subseteq X\}$. The resulting topology is called the *Stone topology* on βX .

We need to check that the Stone base is a topological basis. For each $\mathcal{F} \in \beta X$, there is a Stone set containing it – namely, pick any $A \in \mathcal{F}$ and then $\hat{A}$ will contain $\mathcal{F}$. If $\mathcal{F}$ belongs to $\hat{A}$ and $\hat{B}$, it means $A, B \in \mathcal{F}$ so $A \cap B \in \mathcal{F}$ whence $\mathcal{F} \in \hat{A \cap B}$ which is a basis element satisfying $\hat{A} \cap \hat{B} \subseteq \hat{A \cap B}$ (check this; in fact equality holds) First we have to show how X is a subspace of βX .

Theorem 5.2.15 *There is an embedding (i.e. homeomorphism onto its image) of X into βX .*

Proof: Define $h : X \rightarrow \beta X$ by $x \mapsto \mathcal{F}_x$ where $\mathcal{F}_x$ is the principal ultrafilter $\mathcal{F}_x := \{A \subseteq X : x \in A\}$. This map is clearly injective. Note that the stone basis contains

$$\{\hat{x}\} = \{\mathcal{F} \in \beta X : \{x\} \in \mathcal{F}\} = \{\mathcal{F}_x\}$$

which means that the image $h(X)$ consists of isolated points. Therefore, the topology of X and $h(X)$ are the same (the discrete topology). $\square$

We will continue to refer to $h(X) \subseteq \beta X$ by X . Next, we show that X is dense in βX .

Theorem 5.2.16 *The closure of X in βX is the entire space βX .*

Proof: Let $F \in \beta X$ be any ultrafilter. We aim to show that every neighborhood U of F in βX intersects X . The topology on βX is generated by sets of the form

$$\hat{A} = \{G \in \beta X : A \in G\},$$

where $A \subseteq X$. Since U is a neighborhood of F , there exists an $\hat{A} \subseteq U$ and $F \in \hat{A}$. By definition, $F \in \hat{A}$ means that $A \in F$. Because F is an ultrafilter on X , and $A \in F$, it follows that A is a nonempty subset of X . Choose any $x \in A$. Since $A \subseteq X$, we have $x \in X$. Furthermore, since $A \subseteq X$ and $F \in \hat{A} \subseteq U$, we see that $x \in X \cap U$. Hence $U \cap X \neq \emptyset$. Since this argument holds for every neighborhood U of F , it follows that F is in the closure of X . As F was arbitrary, the closure of X is the entire space βX . $\square$

To prove that βX is a compactification of X , we check that βX is compact and Hausdorff.

Theorem 5.2.17 *βX is compact Hausdorff.*

Proof: (Hausdorff) Let $\mathcal{F}, \mathcal{G} \in \beta X$ be two different ultrafilters. Pick $A \in \mathcal{F}$ for which $A \notin \mathcal{G}$. Then by Theorem 5.2.12, $X \setminus A \in \mathcal{G}$. Then the basis elements $\hat{A}$ and $\hat{X \setminus A}$ are neighborhoods of $\mathcal{F}$ and $\mathcal{G}$ respectively. They are disjoint because an ultrafilter cannot contain both A and $X \setminus A$ at the same time.

(Compact) Let $\mathcal{U}$ be an open cover of βX . WLOG, we may replace each open set in $\mathcal{U}$ by basis sets whose union is that open set (if we can pick a finite subcover after this replacement, we can also pick a finite subcover from $\mathcal{U}$). So now assume $\{\hat{A}\}_{A \in \mathcal{A}}$ is an open cover of βX .

By a cover, it means that every ultrafilter $\mathcal{F}$ contains some $A \in \mathcal{A}$. By Theorem 5.2.12, it means every ultrafilter $\mathcal{F}$ does not contain $X \setminus A$ for some $A \in \mathcal{A}$. Now, if every finite intersection from the collection $\{X \setminus A\}_{A \in \mathcal{A}}$ is non-empty, then $\{X \setminus A\}_{A \in \mathcal{A}}$ would be a filter subbasis by definition 5.2.10, from which we can generate a filter and subsequently obtain an ultrafilter that contains every $X \setminus A$, a contradiction. Therefore, there must exist $A_1, \dots, A_n \in \mathcal{A}$ such that $(X \setminus A_1) \cap \dots \cap (X \setminus A_n) = \emptyset$. That is, $A_1 \cup \dots \cup A_n = X$.

One can check that for any two subsets $A', A'' \subseteq X$, we have $A' \hat{\cup} A'' = \hat{A}' \cup \hat{A}''$. As such, $\hat{A}_1 \cup \dots \cup \hat{A}_n = \hat{X} = \beta X$ and $\{\hat{A}_1, \dots, \hat{A}_n\}$ is our desired subcover. $\square$

Finally, to show that βX is the Stone-Ćech Compactification of X (up to equivalence), we check that the universal property holds. To do this, we have to take a detour to talk about $\mathcal{F}$ -limits.

Definition 5.2.18 Let X be a set as usual. Let C be a topological space, and $\{c_x\}_{x \in X}$ be a collection of points in C . For any $\mathcal{F}$ a filter on X , we say that $c \in C$ is a $\mathcal{F}$ -limit of $\{c_x\}_{x \in X}$, denoted $\lim_{x \rightarrow \mathcal{F}} c_x$, if for every neighborhood $V \subseteq C$ of c , there exists $A \in \mathcal{F}$ such that $c_x \in V$ for all $x \in A$.

Let us discuss some properties of $\mathcal{F}$ -limits that we will need.

Lemma 5.2.19 *If C is Hausdorff, then $\{c_x\}_{x \in X}$ has at most one $\mathcal{F}$ -limit.*

Proof: Let $c \neq d$ be points in C and U, V be disjoint neighborhoods of c, d respectively. If they are both $\mathcal{F}$ -limits, then let $A, B \in \mathcal{F}$ be such that $c_x \in U$ for all $x \in A$ and $c_x \in V$ for all $x \in B$. This forces $A \cap B = \emptyset$, which is a contradiction. $\square$

Lemma 5.2.20 *If C is compact and $\mathcal{F}$ is an ultrafilter, then $\{c_x\}_{x \in X}$ has at least one $\mathcal{F}$ -limit.*

Proof: First we show that any point $c \in \bigcap_{A \in \mathcal{F}} \overline{\{c_x : x \in A\}}$ must be a $\mathcal{F}$ -limit. Indeed, let V be a neighborhood of c . It suffices to show that $B := \{x \in X : c_x \in V\}$ is in $\mathcal{F}$. Suppose otherwise. Then by Theorem 5.2.12, $X \setminus B \in \mathcal{F}$. But $\{c_x : x \in X \setminus B\} \subseteq C \setminus V$ which is closed so $\overline{\{c_x : x \in X \setminus B\}} \subseteq C \setminus V$ which does not contain c , a contradiction.

Next, it remains to show that $\bigcap_{A \in \mathcal{F}} \overline{\{c_x : x \in A\}}$ is non-empty. For each $A \in \mathcal{F}$, $\overline{\{c_x : x \in A\}}$ is non-empty and closed subset of C , hence compact. To apply the Cantor intersection theorem and finish, we have to check that whenever we have finitely many $A_1, \dots, A_n$, $\bigcap_{i=1}^n \overline{\{c_x : x \in A_i\}}$ is non-empty. Indeed, this follows from the observation that $\bigcap_{i=1}^n A_i$ is non-empty. $\square$

Theorem 5.2.21 *Given any continuous map $f : X \rightarrow C$ where C is compact Hausdorff, there exists a unique extension to a continuous map $g : \beta X \rightarrow C$.*

Proof: Uniqueness is shown in Lemma 5.1.4. For each ultrafilter $\mathcal{F}$, lemmas 5.2.19 and 5.2.20 tell us that $\{f(x)\}_{x \in X}$ has exactly one $\mathcal{F}$ -limit c . Define the map $g : \mathcal{F} \mapsto c$. This map is an extension of f ; indeed, we know $f(y)$ is a $\mathcal{F}_y$ -limit because no matter the neighborhood V of $f(y)$ we pick $A = \{y\}$ and then $f(A) = \{f(y)\} \subseteq V$.

It remains to check continuity of g . Let $\mathcal{F} \in X$ be given, denote $g(\mathcal{F})$ by c and let V a neighborhood of c be given. The goal is to produce a neighborhood U of $\mathcal{F}$ such that $g(U) \subseteq V$. Let V' be a neighborhood of c such that $\overline{V'} \subseteq V$ (use normality to do this). Now, there exists $A \in \mathcal{F}$ such that $\{f(x) : x \in A\} \subseteq V'$. Set $U = \hat{A}$. We need to check that if $A \in \mathcal{G}$, then the $\mathcal{G}$ -limit d lies in V . If it does not, then certainly $d \in C - \overline{V'}$. Let $B \in \mathcal{G}$ be such that $\{f(x) : x \in B\} \subseteq C \setminus \overline{V'}$. Then a point $y \in A \cap B$ satisfies $f(y) \in V'$ and $f(y) \in C \setminus \overline{V'}$, a contradiction. $\square$

5.3 Stone-Čech Compactification of the Natural Numbers $\beta\mathbb{N}$.

Definition 5.3.1 The *Stone-Čech compactification* of $\mathbb{N}$, denoted by $\beta\mathbb{N}$, is the unique (up to homeomorphism) compact Hausdorff space containing $\mathbb{N}$ as a dense subspace, with the following universal property: For any compact Hausdorff space K and any continuous function $f : \mathbb{N} \rightarrow K$, there exists a unique continuous extension $\beta f : \beta\mathbb{N} \rightarrow K$ such that $\beta f|_{\mathbb{N}} = f$. In other words, any continuous function from $\mathbb{N}$ to a compact Hausdorff space can be extended uniquely over $\beta\mathbb{N}$.

Theorem 5.3.2 $\beta\mathbb{N}$ is compact Hausdorff space in which $\mathbb{N}$ is a countable dense set.

Proof: Let $\mathbb{N}$ denote the set of natural numbers equipped with the discrete topology. The Stone-Čech compactification $\beta\mathbb{N}$ is characterized by the property that every continuous function $f : \mathbb{N} \rightarrow K$, where K is a compact Hausdorff space, can be uniquely extended to a continuous function $\beta f : \beta\mathbb{N} \rightarrow K$. This universal extension property is fundamental in defining $\beta\mathbb{N}$.

To construct $\beta\mathbb{N}$, consider the set of all ultrafilters on $\mathbb{N}$. An *ultrafilter* p on $\mathbb{N}$ is a maximal filter—a non-empty collection of subsets of $\mathbb{N}$ satisfying:

- (1) If $A, B \in p$, then $A \cap B \in p$.
- (2) If $A \in p$ and $A \subseteq B \subseteq \mathbb{N}$, then $B \in p$.
- (3) $\emptyset \notin p$.
- (4) For any subset $A \subseteq \mathbb{N}$, either $A \in p$ or $\mathbb{N} \setminus A \in p$.

The *principal ultrafilters* correspond to points in $\mathbb{N}$. For each $n \in \mathbb{N}$, the principal ultrafilter p_n is defined by $p_n = \{A \subseteq \mathbb{N} \mid n \in A\}$. We embed $\mathbb{N}$ into $\beta\mathbb{N}$ via the map $n \mapsto p_n$.

We define the topology on $\beta\mathbb{N}$ by stipulating that a set $U \subseteq \beta\mathbb{N}$ is open if and only if, for every continuous function $f : \mathbb{N} \rightarrow K$ into a compact Hausdorff space K , the preimage under the extended function $\beta f : \beta\mathbb{N} \rightarrow K$ of an open set in K is open in $\beta\mathbb{N}$. The topology is thus the weakest one making all such extended functions βf continuous.

To show that $\beta\mathbb{N}$ is Hausdorff, take any two distinct ultrafilters p and q in $\beta\mathbb{N}$. Since $p \neq q$, there exists a subset $A \subseteq \mathbb{N}$ such that $A \in p$ but $A \notin q$. Define a continuous function $f : \mathbb{N} \rightarrow \{0, 1\}$ by setting $f(n) = 1$ if $n \in A$ and $f(n) = 0$ otherwise. The function f extends uniquely to a continuous function $\beta f : \beta\mathbb{N} \rightarrow \{0, 1\}$. Since $\beta f(p) = 1$ and $\beta f(q) = 0$, the points p and q are separated

by the preimages $(\beta f)^{-1}(1)$ and $(\beta f)^{-1}(0)$, which are disjoint open sets in $\beta\mathbb{N}$. Therefore, $\beta\mathbb{N}$ is Hausdorff.

To establish that $\beta\mathbb{N}$ is compact, consider any open cover $\{U_i\}_{i \in I}$ of $\beta\mathbb{N}$. For each i , there exists a continuous function $f_i : \mathbb{N} \rightarrow [0, 1]$ such that U_i contains the preimage under βf_i of an open set in $[0, 1]$. Since $[0, 1]$ is compact and each βf_i is continuous, the images $\beta f_i(\beta\mathbb{N})$ are compact subsets of $[0, 1]$. By the finite subcover property in $[0, 1]$, and the universal property of $\beta\mathbb{N}$, we can extract a finite subcollection $\{U_{i_1}, \dots, U_{i_n}\}$ that still covers $\beta\mathbb{N}$. Thus, $\beta\mathbb{N}$ is compact.

To demonstrate that $\mathbb{N}$ is dense in $\beta\mathbb{N}$, let U be any non-empty open set in $\beta\mathbb{N}$. There exists a continuous function $f : \mathbb{N} \rightarrow K$ into a compact Hausdorff space K and an open set $V \subseteq K$ such that $U = (\beta f)^{-1}(V)$. Since f is defined on $\mathbb{N}$, and V is open, there exists $n \in \mathbb{N}$ with $f(n) \in V$. Therefore, $n \in U$, indicating that $\mathbb{N}$ intersects every non-empty open set in $\beta\mathbb{N}$. Consequently, $\mathbb{N}$ is dense in $\beta\mathbb{N}$. $\square$

Definition 5.3.3 Let X be a topological space with topology τ . The *weight* of X , denoted $w(X)$, is defined as the smallest cardinality of a basis for τ . Formally,

$$w(X) = \min\{|\mathcal{B}| : \mathcal{B} \subseteq \tau \text{ and } \mathcal{B} \text{ is a basis for } \tau\}.$$

A basis $\mathcal{B} \subseteq \tau$ for X is a collection of open sets such that every open set in τ can be written as a union of elements of $\mathcal{B}$. Thus, $w(X)$ represents the minimal "complexity" of describing all open sets in X by unions of elements from $\mathcal{B}$.

Definition 5.3.4 Let X be a topological space with topology τ . The *density* of X , denoted $d(X)$, is the smallest cardinality of a dense subset of X . Formally,

$$d(X) = \min\{|\mathcal{D}| : \mathcal{D} \subseteq X \text{ and } \overline{\mathcal{D}} = X\}.$$

A subset $\mathcal{D} \subseteq X$ is called *dense* in X if its closure $\overline{\mathcal{D}}$ equals X ; that is, every open set in X has a nonempty intersection with $\mathcal{D}$. The density $d(X)$ thus measures the minimal "size" needed for a subset to be dense in X , giving an idea of how "spread out" points must be to influence the entire space.

Definition 5.3.5 Let X be a topological space and $x \in X$. The *character* of X at a point x , denoted $\chi(x, X)$, is the smallest cardinality of a neighborhood basis at x . Formally,

$$\chi(x, X) = \min\{|\mathcal{N}| : \mathcal{N} \subseteq \tau \text{ is a neighborhood basis at } x\}.$$

A *neighborhood basis* $\mathcal{N}$ at x is a collection of open sets containing x such that for any open set U containing x , there exists $V \in \mathcal{N}$ with $V \subseteq U$. The character of X at x thus reflects the "local complexity" of the topology around x . The *character* of X is then defined as

$$\chi(X) = \sup_{x \in X} \chi(x, X),$$

representing the largest character needed at any point in X , and it provides a measure of the "worst-case" local complexity of the topology.

Theorem 5.3.6 The weight of the Stone-Ćech compactification $\beta\mathbb{N}$, denoted $w(\beta\mathbb{N})$, is $2^{\aleph_0}$, the cardinality of the continuum.

Proof: By definition, the weight of a topological space Y , denoted $w(Y)$, is the smallest cardinality of a basis for its topology. Since $\mathbb{N}$ is countable (i.e., $|\mathbb{N}| = \aleph_0$) and discrete, the Stone-Ćech compactification $\beta\mathbb{N}$ has a much larger structure,

containing all ultrafilters on $\mathbb{N}$. The set of all ultrafilters on $\mathbb{N}$ has cardinality $2^{\aleph_0}$, the continuum. In $\beta\mathbb{N}$, the topology is fine enough that the smallest basis must have a cardinality sufficient to describe neighborhoods for every ultrafilter, thus requiring a basis of size $2^{\aleph_0}$. Therefore, $w(\beta\mathbb{N}) = 2^{\aleph_0}$. $\square$

Theorem 5.3.7 *The density of $\beta\mathbb{N}$, denoted $d(\beta\mathbb{N})$, is $\aleph_0$, the cardinality of the natural numbers.*

Proof: By definition, the density of a topological space Y , denoted $d(Y)$, is the smallest cardinality of a dense subset of Y . In the Stone-Čech compactification $\beta\mathbb{N}$, the natural numbers $\mathbb{N}$ itself is dense. This means that the closure of $\mathbb{N}$ in $\beta\mathbb{N}$ is $\beta\mathbb{N}$, i.e., $\overline{\mathbb{N}} = \beta\mathbb{N}$. Since $\mathbb{N}$ is countable, its cardinality is $\aleph_0$. Thus, $d(\beta\mathbb{N}) = \aleph_0$, as $\mathbb{N}$ is a dense subset of $\beta\mathbb{N}$ with cardinality $\aleph_0$. $\square$

Theorem 5.3.8 *The character of $\beta\mathbb{N}$ at each point in $\beta\mathbb{N} \setminus \mathbb{N}$ (the non-principal ultrafilters), denoted $\chi(\beta\mathbb{N})$, is $2^{\aleph_0}$, the cardinality of the continuum.*

Proof: By definition, the character of a point x in a topological space Y , denoted $\chi(x, Y)$, is the smallest cardinality of a local base at x . The character of the space Y is the supremum of $\chi(x, Y)$ over all points $x \in Y$. For points in $\mathbb{N} \subset \beta\mathbb{N}$, the character $\chi(n, \beta\mathbb{N}) = 1$, as singletons form a local base around each point $n \in \mathbb{N}$. However, for points in $\beta\mathbb{N} \setminus \mathbb{N}$, the character is much larger due to the structure of the space. Each point in $\beta\mathbb{N} \setminus \mathbb{N}$ corresponds to a non-principal ultrafilter on $\mathbb{N}$. Any local base at a non-principal ultrafilter must be large enough to capture all cofinite subsets and subsets related to the ultrafilter's filter base, requiring a collection of neighborhoods with cardinality at least $2^{\aleph_0}$. Therefore, $\chi(x, \beta\mathbb{N}) = 2^{\aleph_0}$ for all $x \in \beta\mathbb{N} \setminus \mathbb{N}$. Since this is the maximum character in $\beta\mathbb{N}$, we conclude $\chi(\beta\mathbb{N}) = 2^{\aleph_0}$. $\square$

Theorem 5.3.9 *$\beta\mathbb{N}$ is separable but its subspace $\mathbb{N}^* = \beta\mathbb{N} \setminus \mathbb{N}$ is not.*

Proof: To show that $\beta\mathbb{N}$ is separable, we need to exhibit a countable dense subset. By construction, $\mathbb{N}$ is embedded as a subset of $\beta\mathbb{N}$, and it is countable. Since $\beta\mathbb{N}$ is defined as the smallest compact Hausdorff space containing $\mathbb{N}$ as a dense subset, it follows that $\mathbb{N}$ is dense in $\beta\mathbb{N}$. Therefore, $\mathbb{N} \subset \beta\mathbb{N}$ forms a countable dense subset, establishing that $\beta\mathbb{N}$ is separable.

We now show that the subspace $\mathbb{N}^*$ is not separable, i.e., it does not contain a countable dense subset. Suppose, for contradiction, that $\mathbb{N}^*$ is separable. Then, there exists a countable dense subset $D \subset \mathbb{N}^*$. Note that $\mathbb{N}^*$ is a compact, Hausdorff, totally disconnected space with no isolated points (i.e., it is a perfect space in which every point is a limit point).

This leads to a contradiction with a fundamental result in topology: a compact Hausdorff space that is totally disconnected and perfect cannot be separable. Thus, it is impossible for $\mathbb{N}^*$ to have a countable dense subset. Therefore, no countable dense subset exists in $\mathbb{N}^*$, and $\mathbb{N}^*$ is not separable. $\square$

Theorem 5.3.10 *Let X be a separable Tychonoff space, and suppose there exists a map $f : \mathbb{N} \rightarrow X$ whose image is dense in X . Then the Stone-Čech compactification βX of X can be realized as a quotient of the Stone-Čech compactification $\beta\mathbb{N}$. More precisely, there exists a continuous surjection $\beta f : \beta\mathbb{N} \rightarrow \beta X$ extending f , and hence $\beta X \cong \beta\mathbb{N}/\sim$, where $\sim$ is the equivalence relation induced by βf .*

Proof: Let X be a separable Tychonoff space. By assumption, there exists a countable dense subset $\{x_n : n \in \mathbb{N}\}$ of X . Consider the discrete space $\mathbb{N}$ and define a map $f : \mathbb{N} \rightarrow X$ by $f(n) = x_n$. Since the domain is discrete, every function is continuous, hence f is continuous. Moreover, since $\{x_n : n \in \mathbb{N}\}$ is dense in X , the image of f is dense in X .

Recall that each Tychonoff space Y admits a Stone-Ćech compactification βY , a compact Hausdorff space into which Y is densely embedded. The space βY has the universal extension property: for every continuous map $g : Y \rightarrow K$ into a compact Hausdorff space K , there exists a unique continuous extension $\beta g : \beta Y \rightarrow K$ such that $\beta g|_Y = g$.

Applying this universal property to $f : \mathbb{N} \rightarrow X$, we obtain a unique continuous extension $\beta f : \beta \mathbb{N} \rightarrow \beta X$ with $\beta f|_{\mathbb{N}} = f$. Our aim is to show that βf is a quotient map. First, observe that since $\mathbb{N}$ is dense in $\beta \mathbb{N}$ and X is dense in βX , the image $\beta f(\beta \mathbb{N})$ is dense in βX . Moreover, as $\beta f(\beta \mathbb{N})$ is the continuous image of a compact space, it is itself compact. Thus $\beta f(\beta \mathbb{N})$ is a dense, compact subspace of the Hausdorff space βX .

We claim that this implies $\beta f(\beta \mathbb{N}) = \beta X$. Suppose, for the sake of contradiction, that there exists $x \in \beta X \setminus \beta f(\beta \mathbb{N})$. Since βX is Hausdorff, we can separate x from $\beta f(\beta \mathbb{N})$ by disjoint open neighborhoods, contradicting the density of $\beta f(\beta \mathbb{N})$. Hence no such x can exist, and thus $\beta f(\beta \mathbb{N}) = \beta X$.

Since βf is a continuous surjection from a compact Hausdorff space $\beta \mathbb{N}$ onto a Hausdorff space βX , it is a quotient map. By definition of a quotient map, the topology on βX is the quotient topology induced by βf . Thus, we have shown that βX is a quotient of $\beta \mathbb{N}$. □

6 APPLICATIONS OF STONE-ĆECH COMPACTIFICATION

6.1 Functional Analysis. In functional analysis, the Stone-Ćech compactification is used to extend bounded continuous functions from a locally compact space to a compact space. Given any Tychonoff space X , the space of bounded continuous functions $C_b(X)$ can be uniquely extended to the compactification βX , the largest compact Hausdorff space into which X can be embedded. This is especially useful in Banach space theory, where extending bounded continuous functions allows for a richer topological structure when dealing with limits, dual spaces, and weak* compactness.

Example 6.1.1 Let $X = \mathbb{R}$, and consider the bounded continuous function

$$g(x) = \sin(x) \in C_b(\mathbb{R}).$$

As $x \rightarrow \infty$, $g(x)$ does not approach any limit, oscillating indefinitely between -1 and 1 . Therefore, $g(x)$ cannot be continuously extended to simpler compactifications such as the one-point compactification $\mathbb{R} \cup \{\infty\}$ or the two-point compactification $\mathbb{R} \cup \{\pm\infty\}$, since any continuous extension would require a well-defined limit at infinity. In contrast, using the Stone-Ćech compactification $\beta \mathbb{R}$, there is a unique continuous extension of $g(x)$ to $\beta \mathbb{R}$. This demonstrates the full power of the Stone-Ćech compactification: every bounded continuous function defined on a Tychonoff space X can be extended to βX . Thus, βX captures all possible limit behaviors, no matter how oscillatory or complicated they may be.

In Banach space theory, the dual space $C_b(X)^*$, which consists of bounded linear functionals on $C_b(X)$, can be understood in terms of βX , where compactness ensures that bounded sets in the dual space are weak* compact.

6.2 Algebra and Semigroups. In algebra, the Stone-Čech compactification is particularly useful in the study of semigroups. A semigroup is a set S equipped with an associative binary operation. When the set S is discrete, its Stone-Čech compactification βS becomes a compact right-topological semigroup, where the semigroup operation extends to the compactification, even for points corresponding to ultrafilters.

Example 6.2.1 For $S = \mathbb{N}$, consider the addition operation. In $\beta\mathbb{N}$, the sum of two ultrafilters $p, q \in \beta\mathbb{N}$ is defined by:

$$A \in p + q \quad \text{if and only if} \quad \{n \in \mathbb{N} : A \setminus n \in q\} \in p.$$

This defines an addition operation on $\beta\mathbb{N}$ that generalizes the usual addition of natural numbers to the entire compactification.

This construction is crucial for proving results like *Hindman's Theorem*, which asserts that for any finite partition of $\mathbb{N}$, there exists an infinite subset $A \subseteq \mathbb{N}$ such that all finite sums of distinct elements from A lie in the same partition cell. The compact semigroup structure of $\beta\mathbb{N}$, combined with ultrafilters, allows for a topological proof of this result.

6.3 Set Theory. The Stone-Čech compactification has deep connections to set theory through its use of ultrafilters, which play a central role in Ramsey theory, partition properties, and infinite combinatorics. In set theory, ultrafilters on $\mathbb{N}$ are viewed as points in the compact space $\beta\mathbb{N}$. This allows us to leverage compactness in $\beta\mathbb{N}$ to derive results about infinite sets and partitions that are difficult to obtain using purely combinatorial methods.

Example 6.3.1 Consider *Ramsey's Theorem*, which states that for any finite partition of the edges of a complete graph on $\mathbb{N}$, there exists an infinite monochromatic complete subgraph. More generally, if $\mathbb{N}$ is finitely colored, there exists an infinite monochromatic subset.

The Stone-Čech compactification $\beta\mathbb{N}$ offers a powerful topological framework for proving this theorem. Given any finite partition of $\mathbb{N}$, the compactness of $\beta\mathbb{N}$ ensures the existence of an ultrafilter $p \in \beta\mathbb{N}$ that "selects" one of the partition cells in a way that guarantees large monochromatic subsets. Specifically, if we partition $\mathbb{N}$ into finitely many sets $A_1, A_2, \dots, A_n$, the compactness of $\beta\mathbb{N}$ implies that there is an ultrafilter $p \in \beta\mathbb{N}$ such that for some i , $A_i \in p$. This means that the set A_i is "large" in the sense that it contains elements from all neighborhoods of the ultrafilter, leading to the conclusion of Ramsey's theorem.

6.4 Topological Dynamics. The Stone-Čech compactification is widely used in topological dynamics, where it allows for the study of recurrence phenomena and long-term behavior in dynamical systems. When a topological space X is not compact, but we wish to study the long-term behavior of continuous transformations on X , we can use the Stone-Čech compactification βX to extend the dynamics to a compact space.

Example 6.4.1 Consider the dynamical system $(\mathbb{Z}, T)$, where $T(x) = x + 1$ is the shift map. Although $\mathbb{Z}$ is not compact, the Stone-Čech compactification $\beta\mathbb{Z}$

is compact, and the transformation T extends naturally to $\beta\mathbb{Z}$. This extended transformation allows us to study the recurrence properties of the system in the compact setting.

By working in $\beta\mathbb{Z}$, we can prove results such as *Furstenberg's Recurrence Theorem*, which is closely related to Szemerédi's theorem in additive combinatorics. Furstenberg's Recurrence Theorem states that for any measure-preserving system $(X, \mathcal{B}, \mu, T)$, and any set $A \in \mathcal{B}$ with positive measure, the set of times $n \in \mathbb{N}$ such that $T^n A \cap A \neq \emptyset$ has positive upper density. This is a deep result about the recurrence of sets under a dynamical system and can be approached using the compactness of the Stone-Čech compactification.

6.5 Number Theory. The Stone-Čech compactification has profound applications in additive number theory, particularly in the study of partition regularity, Ramsey theory, and results concerning arithmetic progressions. The compactness of $\beta\mathbb{N}$ allows for topological methods to be applied to problems traditionally studied with combinatorial techniques.

Example 6.5.1 Van der Waerden's Theorem states that for any partition of $\mathbb{N}$ into finitely many sets, one of these sets contains arbitrarily long arithmetic progressions. A result can be obtained using the Stone-Čech compactification $\beta\mathbb{N}$.

To see this, consider a partition of $\mathbb{N}$ into finitely many sets $A_1, A_2, \dots, A_n$. The space $\beta\mathbb{N}$ is compact, so there exists an ultrafilter $p \in \beta\mathbb{N}$ that “selects” one of the partition cells, say A_i . This means that $A_i \in p$, and hence A_i behaves like a “large” set in the ultrafilter sense. The algebraic structure of $\beta\mathbb{N}$, particularly the fact that ultrafilters behave well under addition, ensures that within A_i , there exist arbitrarily long arithmetic progressions. This provides a topological proof of Van der Waerden's theorem, relying on the compactness and algebraic structure of $\beta\mathbb{N}$.

Example 6.5.2 Szemerédi's Theorem is a far-reaching generalization of Van der Waerden's theorem. It asserts that any subset of $\mathbb{N}$ with positive upper density contains arbitrarily long arithmetic progressions. The Stone-Čech compactification $\beta\mathbb{N}$ provides a framework for proving this result by leveraging ultrafilters and recurrence properties.

In this setting, subsets of $\mathbb{N}$ with positive upper density correspond to ultrafilters in $\beta\mathbb{N}$ that capture recurrence properties. These ultrafilters ensure that within any set of positive density, there exist configurations that mimic arithmetic progressions, leading to a proof of Szemerédi's theorem.

7 OPEN PROBLEMS

Question 7.1 Does every infinite compact Hausdorff space contain either a non-trivial convergent sequence or a copy of $\beta\mathbb{N}$?

Comment: This is also known as Efimov's Problem. A counterexample if it exists is called an *Efimov space* and is defined to be an infinite compact Hausdorff space containing neither a non-trivial convergent sequence nor a copy of $\beta\mathbb{N}$, should such a space exist.

Question 7.2 Is it possible to construct a nontrivial map $f : \mathbb{N}^* \rightarrow \mathbb{N}$ without using additional set-theoretic axioms?

Comment: This open problem explores whether one can define a meaningful (nontrivial) function from the remainder $\mathbb{N}^* = \beta\mathbb{N} \setminus \mathbb{N}$ to $\mathbb{N}$ within standard Zermelo–Fraenkel set theory (ZFC) without needing extra axioms, such as those related to the Axiom of Choice. The inquiry explores the structure of $\mathbb{N}^*$ and the limitations of standard set theory in handling certain mappings within compactifications.

Question 7.3 Can $(\beta\mathbb{N}, +)$ be embedded in $(\mathbb{N}^*, +)$?

Comment: This open problem examines whether the structure of $\beta\mathbb{N}$ under addition can be embedded in $\mathbb{N}^*$, which represents the points added in the Stone–Čech compactification of $\mathbb{N}$. It addresses the algebraic structure within compactifications and whether the semigroup operation extends naturally from $\mathbb{N}$ to $\beta\mathbb{N}$ and its remainder $\mathbb{N}^*$.

Question 7.4 Are there p, q, r and s in $\mathbb{N}^*$ such that $p + q = r \times s$?

Comment: This open problem investigates the algebraic behavior of elements within $\mathbb{N}^*$ under operations such as addition and multiplication. It aims to understand if there exist elements in $\mathbb{N}^*$ that satisfy this specific equation, thus probing the complex structure and potential relationships among elements in the nonstandard part of $\beta\mathbb{N}$.

Question 7.5 [15] What spaces can be embedded in $\beta\mathbb{N}$?

Comment: This open problem seeks to identify which other topological spaces can be embedded within the Stone–Čech compactification $\beta\mathbb{N}$. Understanding which spaces fit within $\beta\mathbb{N}$ provides insights into the universality and versatility of the Stone–Čech compactification in embedding other topological constructs, particularly as it relates to spaces with similar properties or compactifications.

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